Labor Market Power and Spatial Policies

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Abstract

Spatial models typically assume competitive labor markets. We develop a spatial equilibrium model with labor market power and calibrate it to four decades of U.S. data. Wage markdowns have roughly halved since 1980, driven by more establishments and rising mobility across industries. Relaxing housing supply constraints and subsidizing migration draw workers to targeted cities, so establishments enter there and exit elsewhere. Because markdowns respond more to entry where they are high, a model without labor market power overstates the welfare gain of these policies when they target large, competitive cities and understates it when they target concentrated labor markets.

Keywords: Labor Market Power; Spatial Models; Housing Policies; Migration Policies.

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1 Introduction

Labor markets differ substantially across U.S. cities. In some, workers can choose among many potential employers and wages closely reflect marginal products; in others, employer concentration is high and mobility frictions limit workers' outside options. The forces that determine this competition have moved in opposite directions since 1980: migration across cities has declined, workers switch industries more often than they used to, and the number of employers competing in the typical local labor market has grown. Quantitative spatial models typically assume competitive labor markets, which keeps high-dimensional equilibria tractable. This paper quantifies the net effect of these trends on local labor market power, which has roughly halved since 1980, and shows that ignoring labor market power changes the predicted welfare gains of spatial policies by 8 to 33 percent. A model without labor market power overstates the gains from relaxing housing supply constraints and from migration subsidies when these policies target large productive cities, and understates them when they target concentrated labor markets.

We develop a spatial equilibrium model in which wage markdowns arise from three sources: costly migration across locations, costly switching across industries, and imperfect competition among firms within local labor markets. Workers choose where to live and which industries to work in, trading off wages, rents, amenities, and the frictions limiting their ability to relocate or change jobs. Firms hire labor in imperfectly competitive local markets (location-industry pairs), and their labor market power depends on both market concentration and the elasticity of local labor supply. The number of establishments in each location-industry-year is determined by a free-entry condition, and firm productivity adjusts with local employment levels through agglomeration economies.

We bring the model to four decades of U.S. Census data, combined with information on housing markets, firm structure, and worker mobility. We recover migration and industry-switching costs from a BLP procedure based on individuals' discrete choices over local labor markets (Berry, Levinsohn and Pakes 1995, Bayer, Keohane and Timmins 2009). The wage coefficient determines the labor supply elasticity and hence the level of markdowns, and we set it so that the average markdown in the model matches the central estimates of the monopsony literature (Sokolova and Sorensen 2021). The remaining housing and labor market parameters follow from inverting the model's equilibrium conditions to exactly match observed wages and housing prices.

The model reproduces two responses that the calibration does not target, the elasticity of city population to the housing-adjusted real wage and the elasticity of the number of establishments to local employment, both of which lie within the range of published estimates. As an alternative, we estimate the wage coefficient with a novel shift-share instrument for workers' location choices, based on labor demand variation induced by free trade agreements with Canada and Mexico, and we repeat the main exercises under this calibration.

We use the model to decompose the evolution of U.S. wage markdowns since 1980 and to study two spatial policies that are standard in the literature and in policy practice: relaxing

housing supply constraints, following the experiment of Hsieh and Moretti (2019), and subsidizing the relocation of low-income workers, a policy for which the Moving to Opportunity experiment is the closest precedent (Chetty, Hendren and Katz 2016). High costs of living in large cities have been a policy concern for many years (Glaeser and Gyourko 2002), and laws promoting accessory dwelling units (ADU) in tight markets have gained in popularity.[2] Each policy is directed at two alternative sets of cities: large productive cities with thick labor markets (New York, San Francisco, and San Jose), and concentrated labor markets where few employers compete. For each policy, we compare the predictions of the model with those of a version without labor market power calibrated to the same data.

We derive three main results. First, local labor market power has declined since 1980: despite falling spatial mobility, the average wage markdown roughly halved, from 28 percent in 1980 to 15 percent in 2019. A Shapley decomposition attributes 45 percent of the decline to the growth in the number of local establishments, which lowers employer concentration, and 36 percent to falling industry-switching costs, which raise the elasticity of labor supply. Rising migration costs worked in the opposite direction. The decomposition is similar when the wage coefficient is set to the shift-share estimate.

Second, when the policies target the large productive cities, the welfare gain with labor market power is smaller than a model without it predicts, by 8.3 percent for the housing policy and by 11.1 percent for the low-income workers eligible for the migration subsidy. When they instead target concentrated labor markets, the gains are 32.6 and 12.6 percent larger. Both policies draw workers toward the targeted cities and away from the rest of the country, so that establishments enter these cities and exit the rest of the country. Markdowns respond more to entry and exit where they are high. When the policies target the large productive cities, where markdowns are low, entry narrows markdowns there only slightly, while exit widens them in the rest of the country. Workers in the rest of the country lose more from the wider markdowns than workers in the targeted cities gain from the narrower ones, which explains why the welfare gain is smaller than in a model without labor market power. The opposite holds when the policies target concentrated labor markets, where markdowns are high and entry narrows them more than exit widens them elsewhere.

Third, the migration subsidy also makes the labor supply of eligible workers more elastic. By lowering the cost of moving to the targeted cities, it makes these workers less likely to stay in the markets where they work, and markdowns there narrow. This partly offsets the effect of establishment exit, which is concentrated in these same markets. With the number of establishments held fixed, the gain of eligible workers is 8.3 and 12.6 percent larger than a model without labor market power predicts, when the subsidy targets the large productive cities and concentrated labor markets respectively. Under the housing policy, instead, lower rents draw less attached workers into the targeted cities but also make the workers already there more likely to stay, and the labor supply elasticity barely changes.

[2]Recent state legislation in Arizona, Colorado, Massachusetts, Iowa, and Washington prevents local jurisdictions from blocking or constraining ADU construction through owner-occupancy mandates, parking mandates, or aesthetic requirements following the example of California.

When the policies target two further sets of cities with intermediate markdowns, the differences between the two models also lie in between. In all four sets of targeted cities, a model without labor market power overstates the welfare gain when the markdown in the targeted cities is lower than in the rest of the country and understates it when it is higher, by more the larger the difference.

Related Literature Quantitative spatial models have been widely used to study the sorting of workers and firms, and a growing body of work incorporates labor market power into spatial settings (Ahlfeldt, Roth and Seidel 2022, Bagga 2023, Bamford 2021, Datta 2024). These models are often computationally demanding due to their high dimensionality, especially when dynamics are involved (Caliendo, Dvorkin and Parro 2019), and adding labor market power increases this complexity. We show that the welfare gains of two standard spatial policies differ from the predictions of a model without labor market power through two margins, establishment entry and the labor supply elasticity, and that the sign of the difference follows the markdown in the targeted cities relative to the rest of the country. These findings speak to a broader literature on spatial policies and the role of place versus people-based interventions (Gaubert et al. 2025, Giannone et al. 2023, Kline and Moretti 2014).[3]

Labor market power has received renewed interest in recent years (Azar and Marinescu 2024, Manning 2021, Sokolova and Sorensen 2021), including work documenting its substantial variation across space (Azar et al. 2020, Azar, Marinescu and Steinbaum 2022, Rinz et al. 2018). We contribute to this literature by deriving local labor supply elasticities from workers' choices of location and industry, and by quantifying the relative importance of the different sources of labor market power: moving costs, industry-switching costs, and local employer concentration.[4] To our knowledge, this is the first decomposition of U.S. wage markdowns over the last four decades that explicitly incorporates spatial frictions. We also provide a new shift-share estimate of the wage coefficient, based on labor demand shocks from free trade agreements with Canada and Mexico. Relatedly, Berger et al. (2024) decompose the contributions of firm granularity, search frictions, and non-wage amenities to monopsony power in a model without geography, using Norwegian data.[5]

Our estimates imply that most of the markdown decline occurred before 2000, with a slower fall thereafter. Yeh, Macaluso and Hershbein (2022) also find falling markdowns before 2000 but estimate rising markdowns afterwards, using a control-function approach on manufacturing plants from 1977 to 2012, while our estimates cover all industries. Deb et al. (2022), in a structural framework similar to Berger, Herkenhoff and Mongey (2022), find markdowns roughly stable since 2000, in line with our post-2000 pattern.

The remainder of the paper is organized as follows. Section 2 shows motivating facts. In

[3]Papers examining housing policy in a spatial context, particularly regulatory constraints on housing supply in major urban areas, include Ganong and Shoag (2017), Hsieh and Moretti (2019), and Greaney (2026). Similar to our approach, Diamond (2016) estimate workers' preferences for cities using a two-step estimator following Berry, Levinsohn and Pakes (1995).

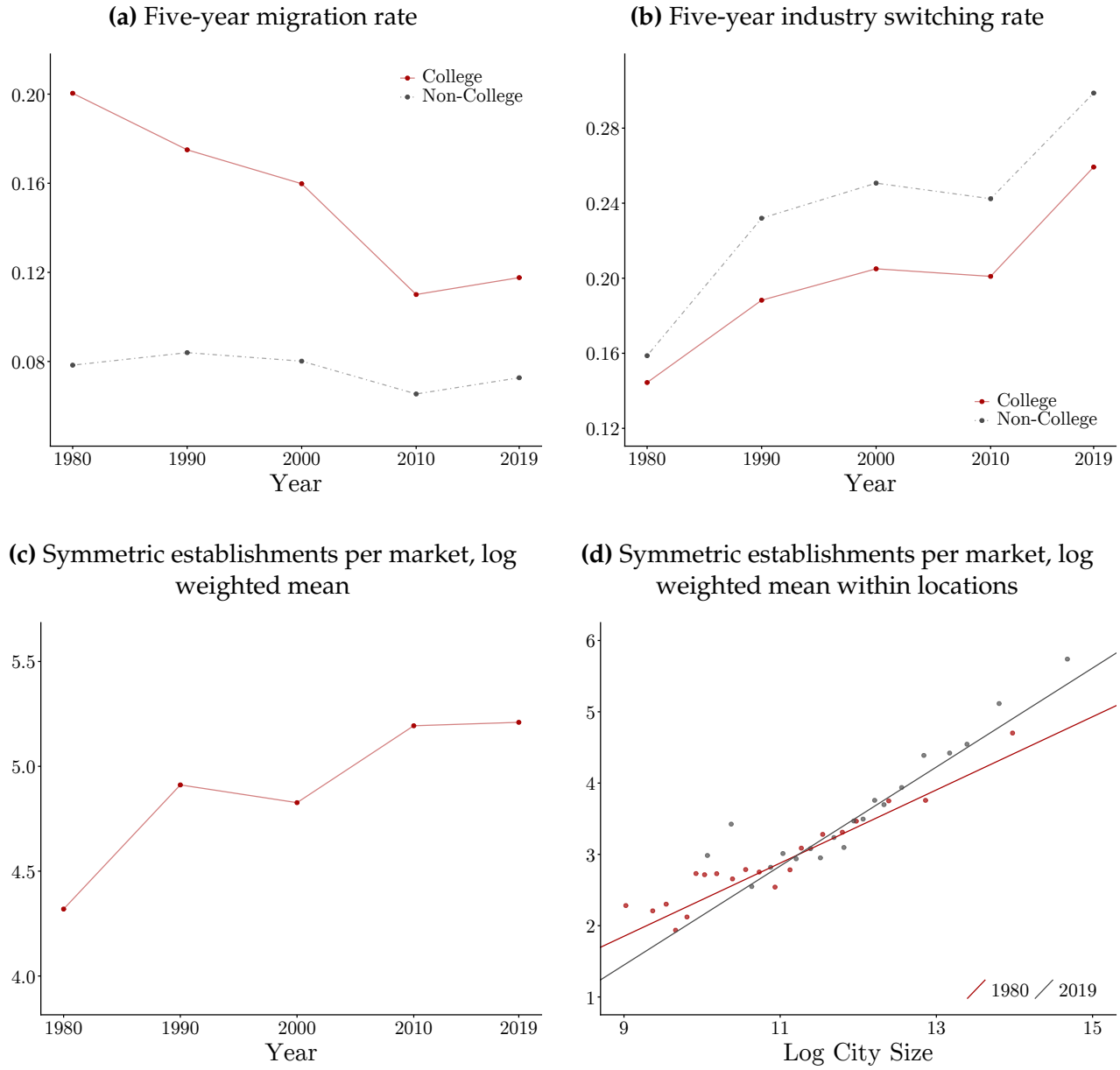
[4]Conceptually similar to our geographic mobility friction, Vial Lecaros, Zárate and Pérez Pérez (2023) and Datta (2024) model commuting costs as a source of local labor market power within metropolitan areas.

[5]Another related paper is Azar, Berry and Marinescu (2022), who estimate the degree of labor market power in the U.S. using a nested logit model.

Section 3, we describe our model framework. Section 4 presents the data and the estimation procedure. In Sections 5 and 6, we report our results and counterfactual exercises. Finally, Section 7 concludes.

2 Facts on Labor Market Power

Figure 1: Trends in potential contributors to labor market power



Notes: Data for Panel (a) are drawn from the U.S. Census (1980–2019); for Panel (b) from the CPS (1975–2017, pooled around each sample year, with 2015–2017 for 2019); for Panels (c) and (d) from the CBP (1980–2016). Panels (a) and (b) report five-year transition rates, recorded directly in the Census through 2000 and constructed from one-year transitions otherwise (Appendix B.1). Panels (c) and (d) plot the log of the employment-weighted mean number of symmetric establishments; in Panel (d), locations are grouped into ventiles of city size and the mean is computed within each ventile. "Symmetric" establishments refer to the hypothetical number of equally-sized establishments that would yield the observed local employment concentration (Adelman 1969).

Figure 1 illustrates how a set of potential contributors to local labor market power have evolved over time. First, five-year migration rates have declined (Panel 1a), especially among college-educated workers. Lower mobility strengthens the spatial segmentation of labor markets and tends to increase monopsony power. Second, job switching across industries has risen sharply (Panel 1b), with similar trends in small and large locations (Appendix Figure F3). Greater cross-industry mobility increases the effective set of employment options available to workers within a given location, reducing firms' ability to exercise labor market power. Third, the number of establishments per location–industry has increased (Panel 1c).[6] More local employer competition also pushes markdowns downward. This growth in establishments reflects two forces. Workers have become increasingly concentrated in larger cities, which host more establishments on average (Panel 1d); moreover, the number of establishments has also risen within location–industry markets (Appendix Figure F1). Taken together, falling migration rates, rising industry switching, and rising establishment counts work in opposite directions, making the overall effect on local labor market power theoretically and quantitatively ambiguous.

Investigating the underlying determinants of these trends lies beyond the scope of this paper. The decline in migration has a large literature.[7] The growth in establishment counts is consistent with the documented divergence between local and national concentration: local concentration has declined since the 1980s even as national concentration rose, for overall economic activity (Rossi-Hansberg, Sarte and Trachter 2021) and for local labor markets (Rinz et al. 2018). The rise in industry switching has received less attention: Kambourov and Manovskii (2008) document rising industry mobility in the PSID through 1997, and our series extends the pattern through 2017.

One possible explanation for the rise in switching among college-educated workers is that industries have become more similar in the tasks they use: the average "task distance" between industries, measured by differences in routine, abstract and manual task content, has fallen by about 7 percent since 1980 for these workers (Appendix Figure F2). These phenomena are treated as inputs in the model and are captured via exogenous time-varying migration and switching costs and fixed costs of entry, except to the extent that they are accounted for by the model's equilibrium outcomes (e.g., differential wage and rent dynamics between large and small cities).

3 Model

We build a model of worker sorting across locations and industries with strategic competition for labor among firms. Workers can move across locations or switch across industries upon paying a utility cost of moving and switching. A finite number of establishments in

[6] We plot "symmetric" establishments (see Section 4.1), the reciprocal of the employment Herfindahl–Hirschman Index (Adelman 1969).

[7] See Jia et al. (2023) for a review; proposed factors include the growing prevalence of dual-earner households, which could reduce family-level mobility due to coordination problems, and lower returns to relocating.

each location-industry produce a homogeneous good using college and non-college labor. In demanding labor, firms act as Cournot players in their local labor markets. How far firms can set wages below the marginal product of labor depends on the number of competing establishments and on the elasticity of the labor supply they face. This elasticity rises with the sensitivity of workers' choices to wages and falls with the costs of moving and switching.

3.1 Workers and Labor Supply

Workers are characterized by their initial location j, industry k, and their educational type $e \in \{n, c\}$. We treat the educational type as a fixed individual characteristic: n and c stand for "non-college educated" and "college educated". Workers choose in which location j' to live and in which industry k' to work. The units of choice are metropolitan areas and three-digit industries. They receive the after-tax wage $y^e_{j'k't}$, which follows from the pre-tax wage $w^e_{j'k't}$ that firms pay through the tax function defined below, and pay rent $r_{j't}$ per unit of housing.

If they choose to move from j to a different location $j' \neq j$, or switch industries $k' \neq k$, they pay the utility cost of moving $\mu^e_t(j, j')$ and switching $\delta^e_t(k, k')$. They derive utility from numeraire consumption c, housing consumption h, and local amenities $\xi^e_{j'k't}$ in their respective locations and industries of choice. Additionally, each worker i receives random utility $\varepsilon_{ij'k't}$ from choosing j' and k'. We assume that $\varepsilon_{ij'k't}$ is independently and identically distributed across workers, location-industry pairs, and years, and follows a Type 1 EV distribution.[8]

$$\begin{aligned} \max_{c,h,j',k'} u_i(c, h, j', k' \mid j, k) \quad &= \quad \beta_c \log(c) + \beta_h \log(h) + \xi^e_{j'k't} - \mu^e_t(j, j') - \delta^e_t(k, k') + \varepsilon_{ij'k't} \\ &\text{s.t.} \quad c + r_{j't} \cdot h \leq y^e_{j'k't}. \end{aligned} \tag{1}$$

Parameter $\mu^e_t(j, j')$ captures the total utility cost of moving and is parameterized as a function of physical distance. Similarly, $\delta^e_t(k, k')$ captures the utility cost of switching industries and is a function of task distances. We allow these costs to vary by education type and over time. Conditional on having chosen location j' and industry k', maximizing the utility function in equation (1) subject to the budget constraint yields that expenditure on housing is a constant share of income:

$$r_{j't} h^* = \frac{\beta_h}{\beta_c + \beta_h} y^e_{j'k't} \equiv \alpha\, y^e_{j'k't}, \tag{2}$$

where α is the housing expenditure share. Plugging optimal housing consumption and optimal private consumption $c^* = \frac{\beta_c}{\beta_c + \beta_h} y^e_{j'k't}$ into utility in equation (1) generates the following indirect utility function

$$\begin{aligned} & v^e_{ij'k't}(y^e_{j'k't}, r_{j't}, \xi^e_{j'k't}, \varepsilon_{ij'k't} | j, k) \\ = \quad & \beta_w \log(y^e_{j'k't}) + \beta_r \log(r_{j't}) + \xi^e_{j'k't} - \mu^e_t(j, j') - \delta^e_t(k, k') + \varepsilon_{ij'k't}, \end{aligned} \tag{3}$$

[8] We normalize the scale of this distribution to one, so the preference parameters and the moving and switching costs are measured relative to the dispersion of the idiosyncratic shocks. Welfare changes expressed in consumption do not depend on this normalization.

where $\beta_w \equiv (\beta_c + \beta_h)$ and $\beta_r \equiv -\beta_h$. Notice that we have omitted inconsequential constants.

Since $\varepsilon_{ij'k't}$ follows a standard Type 1 EV distribution, the probability that a type e individual with initial location j and industry k chooses location j' and industry k' is given by the following expression:

$$p^e_{ij'k't}(j,k) = \frac{\exp\left(\beta_w \log(y^e_{j'k't}) + \beta_r \log(r_{j't}) + \xi^e_{j'k't} - \mu^e_t(j,j') - \delta^e_t(k,k')\right)}{\sum_{j'',k''} \exp\left(\beta_w \log(y^e_{j''k''t}) + \beta_r \log(r_{j''t}) + \xi^e_{j''k''t} - \mu^e_t(j,j'') - \delta^e_t(k,k'')\right)}. \tag{4}$$

Choice probabilities are a function of after-tax wages and rents not only in location j' and industry k' but all other locations and industries. We aggregate these choice probabilities across all individuals of type e and initial location-industries (j,k) to obtain total labor supply N in location j' and industry k':

$$N^e_{j'k't}(y^e_{j'k't}) = \sum_{i=1}^{N^e} p^e_{ij'k't}(j,k). \tag{5}$$

One can then derive the labor supply elasticity (see Appendix Section A.1):

$$\eta^e_{j'k't} = \frac{\partial N^e_{j'k't}}{\partial y^e_{j'k't}} \frac{y^e_{j'k't}}{N^e_{j'k't}} = \frac{\beta_w}{N^e_{j'k't}} \sum_{i=1}^{N^e} p^e_{ij'k't}(1 - p^e_{ij'k't}). \tag{6}$$

The elasticity is proportional to the marginal utility of wages β_w and decreasing in workers' choice probabilities.[9] Labor supply is therefore inelastic in markets whose workers are locked in by high moving and switching costs (p close to one), and approaches β_w in markets that draw their workers from a large pool of individuals, each with a small probability of choosing the market.

Taxes The government raises revenue by taxing labor income to finance public goods, which do not affect individuals' marginal propensity to consume, and to carry out spatial policies (see Section 6). Given mean gross wages $\overline{w}_t$, the average tax rate at gross wage level w^e_{jkt} is given by:

$$T_t(w^e_{jkt}) = 1 - \varsigma_0 \left(\frac{w^e_{jkt}}{\overline{w}_t}\right)^{-\varsigma_1}$$

so that a worker paid the pre-tax wage w^e_{jkt} receives the after-tax wage $y^e_{jkt} = \left(1 - T_t(w^e_{jkt})\right) w^e_{jkt}$.

This formulation of the income tax function is standard in the literature (Bénabou 2002, Guner, Kaygusuz and Ventura 2014). The parameter ς_0 determines the average tax level, while ς_1 determines the progressivity.

[9] Since $N^e_{j'k't} = \sum_i p^e_{ij'k't}$, equation (6) can be written as $\eta^e_{j'k't} = \beta_w \left(1 - \bar{p}^e_{j'k't}\right)$, where $\bar{p}^e_{j'k't} \equiv \sum_i \left(p^e_{ij'k't}\right)^2 / \sum_i p^e_{ij'k't}$ is the employment-weighted average choice probability.

3.2 Firms and Labor Demand

In each location-industry (j,k) there are m_{jkt} symmetric establishments which produce a homogeneous good by combining two types of labor: college-educated n^c and non-college educated n^n. Firm entry into a location-industry is free. Firms are symmetric within location-industries, but differ across location-industries in total factor productivities A_{jkt} and factor shares θ^e_{jkt}. Fixed costs of entry F_{jkt} vary across markets. Due to our assumptions on the workers' discrete choice problem, the labor supply curves in each location-industry are smoothly upward sloping. The higher the wages in a given location-industry, the more workers decide to move in and switch in from other location-industries. Firms engage in Cournot competition for the workers of both types at their local level, taking as given the equilibrium wages in all other location-industries at the national level, and taking as given how worker sorting responds to wages paid in (j,k).[10] The firms' problem is given by

$$\max_{n^c,\, n^n} \; \pi_{jkt} = A_{jkt}\, \mathcal{N}(n^c, n^n) - w^c_{jkt}(n^c, N^c_{jkt})\, n^c - w^n_{jkt}(n^n, N^n_{jkt})\, n^n - F_{jkt}, \tag{7}$$

where $\mathcal{N}(n^c, n^n)$ is a standard CES aggregator of the two types of labor:[11]

$$\mathcal{N}(n^c, n^n) = \left(\theta^c_{jkt} \cdot (n^c)^{\frac{\rho-1}{\rho}} + \theta^n_{jkt} \cdot (n^n)^{\frac{\rho-1}{\rho}}\right)^{\frac{\rho}{\rho-1}}. \tag{8}$$

The inverse labor supply functions $w^e_{jk}(N^e_{jk})$ give the pre-tax wage at which N^e_{jk} workers choose the market, and follow from equation (5) and the tax function. Total labor demand in location-industry (j,k) is the sum across the m_{jkt} establishments, $N^e_{jkt} = m_{jkt}\, n^e$. The extent of firms' local labor market power depends on the number of competitors m_{jkt} and on workers' labor supply elasticity. In particular, wages in each location-industry are given by the education-specific first order conditions:

$$w^e_{jkt} = \underbrace{\frac{m_{jkt}\, \eta^e_{jkt}(1-\varsigma_1)}{1 + m_{jkt}\, \eta^e_{jkt}(1-\varsigma_1)}}_{1\,-\,\text{Markdown}} \text{MPL}^e_{jkt}, \tag{9}$$

where $\text{MPL}^e_{jkt} \equiv A_{jkt}\, \partial\mathcal{N}/\partial n^e$ is the marginal product of labor and η^e_{jkt} is the labor supply elasticity from equation (6). The factor $(1-\varsigma_1)$ converts the workers' elasticity with respect to the after-tax wage into the elasticity firms face when setting pre-tax wages (Appendix A.1). With perfect competition, wages are equal to the marginal product of labor and markdowns

[10] Product markets remain competitive. Product market power would operate mainly through markups on locally sold output, moving profits and establishment entry, without directly changing the labor supply elasticity that determines the markdown. In the model of Kroft et al. (2025), who estimate both forms of market power in the US construction industry, markups add a separate wedge on top of the markdown: for the labor supply elasticity they estimate, the wage is 20 percent below the value of the marginal product of labor without markups and more than 30 percent below it with them. Because we infer productivity from wages assuming competitive product markets, our markdown corresponds to the smaller gap. A shift-share estimate of the wage coefficient β_w on the subsample of traded industries, whose output prices are largely set outside the local market, is close to the all-industry one (Appendix D.4).

[11] We restrict $\theta^c_{jkt} + \theta^n_{jkt} = 1$, as is typical.

are zero.[12] This happens if either the number of establishments is arbitrarily large ($m_{jkt} \to \infty$) or if labor supply is perfectly elastic ($\eta^e_{jkt} \to \infty$), which occurs when moving and switching costs are negligible or when the marginal utility of wages, β_w, is very high.

The number of establishments m_{jkt} per location-industry-year is determined by a free-entry condition specific to each market. In the counterfactual exercises, we allow the productivity of firms to adjust with local employment sizes through agglomeration economies (Duranton and Puga 2004). These agglomeration economies are not internalized by firms and are assumed to follow the log-linear relationship

$$\log A_{jkt} = a_{jkt} + \lambda \log \textstyle\sum_{k''} (N^n_{jk''t} + N^c_{jk''t}), \tag{10}$$

where λ, the elasticity of productivity to city employment, captures the strength of agglomeration economies. Because free entry requires $\pi^*_{jkt} = F_{jkt}$, any counterfactual change in productivity, wages, or employment that raises variable profits induces additional firm entry until profits return to the exogenous entry cost. Symmetry keeps the entry margin tractable: with heterogeneous firms, the within-market size distribution would become a state variable of the equilibrium. Measuring establishments as symmetric equivalents (Section 4.1) preserves the observed concentration of each local labor market, but with heterogeneous firms the response of entry to a policy would also depend on the size of entrants relative to the establishments already in the market. Appendix C.6 compares the model's entry response with cross-sectional evidence and published estimates.

3.3 Housing Markets

Housing demand h^D in location j equals the sum of housing demand of workers in location j, aggregated across industries k:

$$h^D_{jt} = \sum_k N^n_{jkt} \frac{-\beta_r y^n_{jkt}}{\beta_w r_{jt}} + \sum_k N^c_{jkt} \frac{-\beta_r y^c_{jkt}}{\beta_w r_{jt}}. \tag{11}$$

We model housing supply as upward sloping with location-specific constant elasticities and intercepts, following Saiz (2010), with housing measured relative to its 1980 level $h_{j,1980}$:

$$r_{jt} = \kappa_{jt} \left(\frac{h^S_{jt}}{h_{j,1980}} \right)^{\gamma_{jt}} \tag{12}$$

The intercept κ_{jt} may reflect location-specific construction costs, whereas variation in the inverse housing supply elasticity γ_{jt} could stem from differences in developable land and local regulatory constraints. The normalization adapts Greaney (2026), who measures housing relative to its 1970 level; we use 1980, the first year of our data. It leaves the benchmark equilibrium unchanged, since κ_{jt} is calibrated to observed rents, and makes the effect of a policy

[12] Throughout, the markdown is the proportional gap between the marginal product of labor and the wage, $\psi^e_{jkt} \equiv (\text{MPL}^e_{jkt} - w^e_{jkt})/\text{MPL}^e_{jkt}$, so a decline in markdowns over time reflects a decline in labor market power.

that lowers γ_{jt} independent of the units in which housing is measured.

3.4 Equilibrium

An equilibrium is a distribution of workers across locations and industries, sets of wages and rents, and tax rates such that local labor markets clear for both types of labor, firms' labor demand is an equilibrium of the augmented Cournot game within each local labor market, housing markets clear, the distribution of workers across locations and industries reproduces itself under the equilibrium choice probabilities, and the government budget is balanced. We provide a full definition in Appendix Section A.3.

4 Estimation

Our estimation proceeds in three steps. First, we recover migration and industry-switching costs using a BLP procedure based on individuals' discrete choices over local labor markets (Berry, Levinsohn and Pakes 1995, Bayer, Keohane and Timmins 2009). Second, we set the marginal utility of wages, which determines the labor supply elasticity and hence the level of markdowns, so that the average markdown in the model matches the monopsony literature, and tie the marginal utility of rents to it through the housing expenditure share. Finally, we recover the remaining housing and labor-market parameters by inverting the model's equilibrium conditions to exactly match observed wages and housing prices, and internally calibrate unobserved amenities to replicate the allocation of workers across locations and industries. A shift-share estimate of the marginal utility of wages gives an alternative calibration (Appendix D).

4.1 Data

In this section, we document the data sources used for the model inputs. Further details are provided in Appendix Section B.1. Table 1 reports summary statistics for all the listed variables.

Labor Supply We use the 5 percent samples of the US Census 1980-2000 and the ACS 5-year samples from 2007-2011 and 2015-2019 for information on individual location choices, industry choices, wages, rents, and education.[13] Locations are defined as a metropolitan statistical area, and industries follow the 3-digit Census industry classification.[14]

Movers are defined as individuals who have moved across locations during the last five years, the horizon at which the mobility parameters are estimated. For movers, we define the

[13]We focus on full-time employed individuals aged 20 to 65, earning non-zero wages, who are not institutionalized and not in the military. We group individuals by educational attainment "college degree" versus "no college degree".

[14]The sample covers 219 metropolitan areas. In 2019, the public-use geography of the ACS prevents identifying 16 small metropolitan areas, whose residents cannot be assigned to their metropolitan area from the available PUMA boundaries, leaving 203 locations in that year. These areas account for about 1.3 percent of the metropolitan population.

location in which they lived five years ago as their initial location and the current location as the observed optimal choice. For non-movers, the initial location and observed optimal location coincide.[15] The Census records the location of residence five years prior directly through 2000; for 2011 and 2019, we construct five-year migration rates from the one-year rates recorded in the ACS (Appendix B.1).

Industry switchers are individuals who have switched jobs to another 3-digit industry over the same five-year horizon. While the current industry of individual employment is recorded in the Census and ACS data, there is no information on previous industries in which individuals might have worked. We therefore use information from the Current Population Survey (CPS) 1975-2017 to observe one-year switching choices across 3-digit Census industries.[16] We convert the one-year flow tables to a five-year equivalent using transition-probability ratios computed from the Panel Study of Income Dynamics, which follows the same workers over long horizons (Appendix B.1). Jointly, locations and industries constitute the choice set for workers in our model, and previous location, previous industry, and educational group constitute the state variables. Since we focus on location and industry choices within a five-year window, we treat educational attainment as a fixed individual characteristic.

Wages in each location-industry are measured as fixed effects per location-industry, conditional on a set of individual observables to eliminate potential confounding from differences in demographic composition. Location-level wages, required for the estimation procedure described in Section 4.2, are obtained as location fixed effects conditional on industry fixed effects. Similarly, industry-level wages are estimated conditional on location fixed effects. Rents in each location are measured as the annualized user cost of housing, conditional on housing characteristics. We follow the literature closely in constructing these, and hence relegate the details to Appendix Section B.1.

Labor Demand To quantify the degree of Cournot competition on the labor demand side, we compute the number of employers per location–industry as "symmetric" equivalents, following Adelman (1969). This represents the hypothetical number of equally-sized establishments that would generate the observed local employment concentration levels, and is obtained as the reciprocal of the employment Herfindahl–Hirschman Index (HHI). Formally, the symmetric number of establishments in each market is defined as

$$m_{jkt} = \frac{1}{\text{HHI}_{jkt}} = \left(\sum_{l=1}^{M_{jkt}} s_{ljkt}^2 \right)^{-1} \tag{13}$$

where s_{ljkt} denotes the employment share of establishment l in location–industry (j, k) at time t, within a market containing M_{jkt} (potentially asymmetric) establishments.

We measure m_{jkt} using the County Business Patterns (CBP) 1980-2016, an annual series

[15] We drop individuals who lived abroad before their move.

[16] We pool multiple years to increase sample size: 1975–1984 for 1980, 1985–1994 for 1990, 1995–2004 for 2000, 2005–2014 for 2011, and 2015–2017 for 2019.

Table 1: Summary statistics

	Observations	Mean	SD	Min	Max
Prices					
ln Non-college wage	111,003	10.621	0.277	8.866	13.289
ln College wage	111,003	11.037	0.422	8.674	13.660
ln Rent	111,003	9.654	0.246	9.017	10.631
Labor Demand					
Establishments (symmetric)	111,003	24.052	90.826	1.000	6,949.431
Labor Supply: Choice set					
Locations	219				
Industries	177				
Location-industries	31,635				
Task distances non-college	31,329	1.592	0.835	0.000	5.589
Task distances college	31,329	1.279	0.820	0.000	6.433
Labor Supply: Choices					
Five-year moving non-college	1,079	0.087	0.051	0.000	0.487
Five-year moving college	1,079	0.191	0.101	0.000	0.633
Five-year switching non-college	885	0.233	0.079	0.034	0.496
Five-year switching college	885	0.229	0.098	0.000	0.664
Housing Supply					
Elasticity	1,078	2.235	1.184	0.595	7.842
Share land unavailable	1,078	0.272	0.198	0.004	0.883

Notes: Data sources include U.S. Census 1980, 1990, 2000, ACS 2007-2011, 2015-2019, CPS 1975-2017, and DOT 1977. The choice set includes 219 metropolitan areas (203 in 2019; see text). Housing cost annualized for owners. Establishment counts are unweighted averages of symmetric-equivalent establishments per location-industry, based on CBP 1980-2016 data. Moving and switching rates are five-year rates, constructed as described in the text. Housing supply elasticities are drawn from Saiz (2010), and land unavailability data from Lutz and Sand (2023).

published by the U.S. Census Bureau that reports the number of establishments with paid employees, disaggregated by county, industry, and employment-size class (twelve bins ranging from 1–4 to 5000+ workers). We assign each establishment the midpoint of its employment-size interval, and map counties to their corresponding metropolitan areas to compute the Herfindahl–Hirschman Index (HHI) by industry–location for each year. Because data for smaller counties are heavily censored after 2017, we use 2016 data to represent the 2019 decade. The number of symmetric establishments is then computed according to equation (13).

Task Distance Cross-industry switching costs depend on the task distance between industries, which we measure as the Euclidean distance between industry task vectors: following Autor and Dorn (2013), we use the Dictionary of Occupational Titles (DOT 1977) to score each occupation on routine, abstract, and manual task content, and aggregate to the industry level with decade-specific employment weights. In 2019, for example, the distance between banking and credit agencies is 0.09 for college-educated workers, while the distance between trucking and banking is 4.7 for non-college workers. The employment-weighted median distance across industry pairs is 1.7 for non-college and 1.8 for college-educated workers.

Housing Supply Metropolitan area–specific estimates of housing supply elasticities are taken from Saiz (2010). We complement these with county-level data on the share of land unavailable for development from Lutz and Sand (2023), which we aggregate to the location

level using population-weighted averages.

4.2 Mobility Costs

We estimate the costs of moving across locations and of switching across industries from observed flows, with a BLP procedure based on individuals' discrete choices (Berry, Levinsohn and Pakes 1995, Bayer, Keohane and Timmins 2009). The location and industry choices are estimated in separate steps, because the data do not allow us to jointly observe current and previous location-industry choices.[17] Industry-switching rates follow similar trends in small and large locations (Appendix Figure F3), which supports treating the two choices separately.

The observed location choice shares by educational type and initial location are matched to choice probabilities from the model to estimate the mean utility of each location, v^e_{jt}, and the moving costs $\mu^e_t(j, j')$. Given the Type I EV distribution of individual idiosyncratic preferences, the model choice probabilities are given by:

$$p^e_t(j' \mid j) \;=\; \frac{\exp(\overbrace{\beta_w \log(y^e_{j't}) + \beta_r \log(r_{j't}) + \xi^e_{j't}}^{=v^e_{j't}} - \mu^e_t(j, j'))}{\sum_{j''} \exp\left(\beta_w \log(y^e_{j''t}) + \beta_r \log(r_{j''t}) + \xi^e_{j''t} - \mu^e_t(j, j'')\right)} \tag{14}$$

The moving cost is a fixed cost paid for any move plus increments for moves longer than 100, 500, 1,000, and 2,000 kilometers, and it varies across educational types and years. We estimate it separately for each educational type and year from the bilateral flows of workers between locations, by maximum likelihood with a nested fixed point (Berry, Levinsohn and Pakes 1995, Bayer, Keohane and Timmins 2009, Mathes 2025). For given moving costs, a contraction mapping finds the mean utilities v^e_{jt} that reproduce each location's observed share of workers; the likelihood then selects the moving costs that best fit the flows between each pair of locations. Because the mean utilities absorb everything that makes a location attractive to all workers, the fixed cost is identified by the share of workers who stay in their location, and the distance increments by how flows decline with distance. Switching costs are estimated analogously from industry flows, as a function of the task distance between industries, with increments for distances above thresholds between 0.05 and 4. Appendix B.3 gives the cost functions and the likelihood, and Appendix Tables F1 and F2 report the estimates.

Given the preference and cost parameters, we invert the amenities ξ^e_{jkt} in equation (4) through a fixed-point iteration, so that the model reproduces the observed allocation of workers across location-industry cells, given the location-industry wage fixed effects (Appendix Section B.1).[18] The joint model restricts both choices to a single wage coefficient, so a worker

[17] Consistent with the separation in the estimation procedure, location-level wages used in the migration choice are estimated controlling for industry fixed effects, while industry-level wages used in the industry-switching choice are estimated controlling for location fixed effects (see Appendix Section B.1). Even if current and previous location-industry choices were jointly observed, solving a system with $219 \times 177 = 38{,}763$ possible states and choices would be computationally challenging.

[18] Evaluated directly, equation (4) requires the relocation cost between every ordered pair of occupied location-industry cells, a matrix of about 25,000 by 25,000 that the iteration would rebuild at every step. Moving costs

values a wage difference identically whether it arises from moving to a higher-wage city or from switching to a higher-wage industry, and ξ^e_{jkt} absorbs any interaction between the two margins that this restriction misses.[19]

4.3 Parameters

Table 2 summarizes the model parameters. Some parameters are drawn from the literature: the CES production elasticity ρ (Card 2009), the inverse housing supply elasticities γ_{jt} (Saiz 2010), the tax function parameters $(\varsigma_0, \varsigma_1)$ (Guner, Kaygusuz and Ventura 2014), the agglomeration elasticity λ, and the housing expenditure share α. Migration costs $\mu^e_t(j, j')$ and industry switching costs $\delta^e_t(k, k')$ are estimated from observed migration and industry-switching flows (Section 4.2); Appendix Figure F4 shows the estimated migration-cost schedules for each sample year. The wage coefficient β_w is calibrated as described below, with $\beta_r = -\alpha\beta_w$. The productivity parameters $(A_{jkt}, \theta^n_{jkt}, \theta^c_{jkt})$ and housing supply intercepts κ_{jt} are chosen to exactly replicate equilibrium wages and housing prices in the data, while the amenity terms ξ^e_{jkt} are internally calibrated to match the observed allocation of labor across local labor markets.

Table 2: Model parameters

	Parameter	Value	Source
External Parameters:			
CES production function parameter	ρ	2	Card (2009)
Inverse housing supply elasticity	γ_{jt}	Table 1	Saiz (2010)
Tax function	$(\varsigma_0, \varsigma_1)$	(0.902, 0.036)	Guner, Kaygusuz and Ventura (2014)
Agglomeration elasticity	λ	0.04	Melo, Graham and Noland (2009)
Housing expenditure share	α	0.30	Hsieh and Moretti (2019)
Estimated:			
Migration costs	$\mu^e_t(j, j')$	Table F1	Migration flows
Industry switching costs	$\delta^e_t(k, k')$	Table F2	Switching flows
Calibrated:			
Indirect utility parameters	(β_w, β_r)	(0.726, −0.218)	Mean markdown
Agglomeration constants	a_{jkt}	Table F3	Productivity
Productivity	$(A_{jkt}, \theta^n_{jkt}, \theta^c_{jkt})$	Table F3	Wages
Housing supply intercepts	κ_{jt}	Table F3	Housing prices
Amenities	ξ^n_{jkt}, ξ^c_{jkt}	Table F3	Population

Notes: Under the shift-share calibration of Appendix D, $(\beta_w, \beta_r) = (5.087, -1.526)$.

Wage Coefficient We set the wage coefficient β_w to match the firm-level labor supply elasticity estimated in the monopsony literature.[20] Among the studies in the meta-analysis of

depend only on the two locations and switching costs only on the two industries, so the choice probabilities factor exactly into products of a location matrix and an industry matrix, and the large matrix is never formed. We iterate until the model's allocation matches the data to 10^{-8} of total population, the same tolerance to which we solve the counterfactuals.

[19] This interaction is small in U.S. data: in linked employer-employee records, place and industry pay premia are close to additively separable, and place-specific industry returns account for only a few percent of the variance of place premia (Card, Rothstein and Yi 2025).

[20] The wage coefficient is the only parameter that changes markdowns without changing the fit to observed migration and industry switching. Mobility costs are estimated from observed flows before wages enter (Section 4.2). In the benchmark, the amenities ξ^e_{jkt} adjust so that the model reproduces the observed allocation at any

Sokolova and Sorensen (2021) published in leading journals, the average estimate, weighting each study equally, is 4.5, which implies a markdown of about 18 percent (Kline 2025). We choose β_w so that the employment-weighted mean markdown in the model, pooled over location-industry cells and the five sample years, equals 18 percent, which gives $\beta_w = 0.73$.[21] The rent coefficient is tied to the wage coefficient through the housing expenditure share, which we set to $\alpha = 0.30$, in line with the 0.32 that Hsieh and Moretti (2019) adopt from expenditure data, so that $\beta_r = -\alpha\beta_w$.[22]

As an alternative, we provide a new estimate of β_w, using a shift-share instrument that exploits labor demand variation induced by free trade agreements with Canada and Mexico (Appendix D). The estimate is 5.09, with an exposure-robust standard error of 3.01. It implies a firm-level labor supply elasticity of about 22 and an employment-weighted mean markdown that falls from 8 percent in 1980 to 3 percent in 2019. The instrument relates twenty-year changes in wages and rents to migration over five-year windows, so it measures a long-run response, while many estimates in the monopsony literature measure responses over months or quarters. In a dynamic monopsony model, the long-run elasticity equals the short-run elasticity divided by the quit rate; with turnover rates of about 20 percent, Manning (2011) suggests multiplying short-run estimates by 5, which turns the average of 4.5 into a long-run elasticity of about 22. This value is close to the shift-share estimate and to the average of 22.8 among the studies in Sokolova and Sorensen (2021) that identify the elasticity from the inverse labor supply elasticity. The shift-share estimate therefore gives an upper bound on the elasticity, and a lower bound on markdowns.

Under the shift-share calibration, the decomposition of the markdown decline is similar, and so is the difference between the welfare gains of the policies with and without labor market power when the number of establishments is held fixed (Appendix D).[23]

Agglomeration Elasticity We set the agglomeration elasticity to $\lambda = 0.04$, within the range of the median estimates that Melo, Graham and Noland (2009) report across dependent variables in their meta-analysis of 34 studies (0.032 to 0.057), and close to their median of 0.036 for the United States.[24]

value of β_w, and the choice probabilities therefore do not depend on it. The labor supply elasticity in equation (6) is then proportional to β_w. Higher mobility costs would also widen markdowns, but the model would no longer reproduce the observed flows.

[21] The firm-level elasticity is the labor supply elasticity faced by a single establishment when setting its pre-tax wage, $m_{jkt}\,\eta^e_{jkt}(1-\varsigma_1)$: under symmetric Cournot competition each establishment employs a share $1/m_{jkt}$ of its market, so the elasticity of its residual labor supply is m_{jkt} times the market elasticity, and the factor $(1-\varsigma_1)$ converts the after-tax elasticity into the pre-tax one. The markdown in equation (9) equals one over one plus this elasticity, so an elasticity of 4.5 corresponds to a markdown of $1/5.5 \approx 0.18$. Because the elasticity is unbounded above, its mean across cells is dominated by the thickest markets; we therefore target the mean markdown, which aggregates the same information over a bounded object. Allowing the firm to internalize the effect of its wage on local rents barely changes the elasticity (Appendix A.2).

[22] From the indirect utility in equation (3), $\beta_w = \beta_c + \beta_h$ and $\beta_r = -\beta_h$. With Cobb–Douglas preferences the housing expenditure share is $\alpha = \beta_h/(\beta_c + \beta_h)$ (equation (2)), so $\beta_r = -\alpha\beta_w$.

[23] At this elasticity, free entry is not well defined in some markets, so we compare the policies with the number of establishments held fixed.

[24] These estimates assume competitive labor markets, under which the elasticities of wages and of productivity to city size coincide.

Productivity, Housing, and Entry Costs Given the labor supply parameters, we recover the implied distribution of labor supply elasticities across education groups and local labor markets (equation 6). Using the observed equilibrium allocation of workers across industries and locations, the firms' first-order condition in equation (9) and the housing supply function in equation (12) then link observed wages and rents to the productivity and housing supply parameters. We invert these equilibrium conditions to recover the housing supply intercepts κ_{jt} and productivity parameters θ^c_{jkt} and A_{jkt} that make model wages and housing prices exactly match their empirical counterparts in each location–industry cell (Appendix Section B.2). Given the agglomeration elasticity λ, the agglomeration constants a_{jkt} follow from equation (10). The fixed costs of entry are recovered from the free-entry condition: we set F_{jkt} so that $\pi^*_{jkt} = F_{jkt}$ holds at the observed establishment count. In the counterfactual exercises, entry responses are movements in m_{jkt} at fixed F_{jkt}.

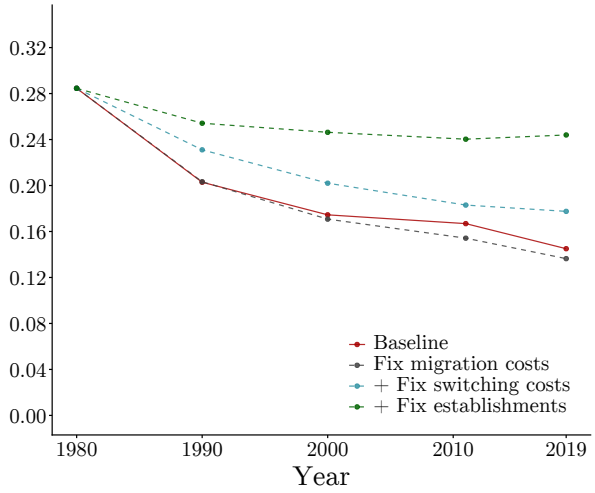
Untargeted Moments We compare two responses that the calibration does not target with published estimates, one for each margin on which the counterfactuals rely. The first is the response of population to the real wage. The model estimates migration and switching costs from observed flows and holds them fixed in the counterfactuals. This is a potential source of misspecification, because the cost of moving depends on choices that housing and migration policies can change, such as housing tenure, including homeownership and coresidence with parents (Luccioletti 2024), and the joint location decisions of couples (Mincer 1978, Costa and Kahn 2000). If these choices respond to local prices, a model with exogenous mobility costs, which abstracts from them, would predict a different response of population to those prices than the one estimated in the data. In the model, the elasticity of city population to the housing-adjusted real wage is 1.2, close to the 1.12 estimated by Suárez Serrato and Zidar (2016) and to the value of about 1.2 implied by the estimates of Allcott and Keniston (2018).

The second is the response of the number of establishments to local employment. Establishment entry is the margin through which labor market power changes the welfare effect of the housing policy (Section 6), and the model represents it with symmetric establishments that enter until operating profits cover a fixed cost. The fixed costs equal the operating profit of each establishment at the observed number of establishments, and the agglomeration elasticity comes from the literature, so the calibration does not target the response of entry to employment. More workers raise the operating profit of each establishment, both directly and because agglomeration raises productivity. Counted as in the data rather than as symmetric employers, the number of establishments rises by 0.79 percent for each percent increase in local employment, within the range of published estimates: 0.26 for immigration into U.S. commuting zones (Mahajan 2024) and between 0.57 and 1.71 for the settlement of refugees in post-war West Germany (Peters 2022) (Appendix C.6 and Table C5).

5 Labor Market Power Over Time

Given the calibrated model, we compute markdowns for each local labor market using equation (9). Figure 2 displays the employment-weighted mean markdown across location-industry cells for each decade from 1980 to 2019 (solid red line). Labor market power has declined substantially: under the main calibration, the mean markdown falls from 28.5 percent in 1980 to 14.5 percent in 2019, roughly halving. Most of the decline occurred before 2000, when the mean markdown was 17 percent.[25]

Figure 2: Mean markdowns across location-industries



Notes: The figure plots the employment-weighted mean markdown across location-industry cells by decade. The red solid line shows the model with all parameters at observed-year values. Dashed lines report a cumulative decomposition in which parameters are progressively fixed at their 1980 values: first migration costs, then additionally industry-switching costs, then additionally the number of establishments per location-industry. Each line corresponds to 2019 conditions with 1980 values for the indicated channels; in 1980 all lines coincide by construction. The plotted cases are a subset of the eight combinations underlying the Shapley attribution in Table 3. Wages, rents, and populations are held at actual-year values, and unobserved utilities are recalibrated so that the observed allocation remains a fixed point.

To understand the mechanisms behind this decline, we solve the model under every combination of three channels held at their 1980 values: migration costs, industry-switching costs, and the number of establishments per location-industry, eight cases in total, and attribute the explained portion of the decline to the three channels using their Shapley values.[26] The decomposition holds wages, rents, and populations at their observed values and asks which channels account for the markdown decline mechanically.

Fixing all three channels at their 1980 values, the 2019 mean markdown would be 24.4 percent rather than the observed 14.5 percent, so the three channels account for 71 percent of the decline. In the Shapley attribution, establishment growth accounts for 45 percent. The

[25] Aggregating cell markdowns with payroll weights instead of employment weights, following Berger, Herkenhoff and Mongey (2022), gives a nearly identical series, with the mean markdown falling from 28.9 percent in 1980 to 14.6 percent in 2019. Weighting cells by employment, the median markdown falls from 20.8 percent in 1980 to 8.5 percent in 2019, and the 90th percentile from 67.3 to 37.4 percent.

[26] The Shapley value averages each channel's marginal contribution over all orderings, so the attribution does not depend on the ordering choice inherent in a cumulative sequence of counterfactuals. Figure 2 plots one cumulative path for readability.

number of establishments per local labor market grew substantially over 1980–2019 (Figure 1c), lowering employer concentration. Along the cumulative path of Figure 2, with migration and switching costs already at their 1980 values, additionally fixing establishment counts raises the 2019 mean markdown from 17.8 to 24.4 percent. In the Shapley attribution, falling industry-switching costs account for a further 36 percent. Workers switch industries more often than in 1980 (Figure 1b), which the model attributes to declining switching costs; along the same path, additionally fixing switching costs raises the 2019 mean markdown from 13.6 to 17.8 percent.

Migration costs moved in the opposite direction. Observed migration rates declined from 1980 to 2019, which the model attributes to higher estimated migration costs, in particular for longer moves (Appendix Table F1). Along the cumulative path, fixing migration costs at their 1980 values lowers the 2019 mean markdown from 14.5 to 13.6 percent. Averaged over orderings, rising migration costs raise the 2019 markdown, and their Shapley value offsets 10 percent of the decline.

General Equilibrium and Shift-Share Calibration The decomposition is similar when prices, populations, and establishment counts adjust in general equilibrium: the three channels account for 65 percent of the decline, against 71 percent with prices held at their observed values, and establishment growth and falling switching costs remain the two largest (Table 3 and Appendix E).

The decomposition is also similar under the shift-share calibration, which sets the wage coefficient to the estimate from the trade-agreement instrument (Section 4.3): establishment growth and falling switching costs account for 34 and 31 percent of the decline, against 45 and 36 percent under the main calibration (Table 3 and Appendix D).

Table 3: Decomposing the decline in markdowns

	Mean markdown			Contribution to the decline (%)			
	1980	2019	Counterfactual	Establishments	Switching	Migration	Explained (%)
Main calibration							
Partial equilibrium	0.285	0.145	0.244	45	36	−10	71
General equilibrium			0.236	45	28	−8	65
Shift-share calibration							
Partial equilibrium	0.082	0.030	0.060	34	31	−9	57
General equilibrium			0.053	29	20	−6	44

Notes: Employment-weighted mean markdowns across location-industry cells. The counterfactual is the 2019 mean markdown with all three channels at their 1980 values; the explained decline is its difference from the observed 2019 markdown, as a percent of the 1980–2019 decline. Each Shapley value averages a channel's contribution over all eight combinations of the three channels held at 1980 values, as a percent of the 1980–2019 decline. In partial equilibrium, wages, rents, and populations are held at observed values; in general equilibrium, they and establishment counts adjust.

6 Labor Market Power and Spatial Policies

We study two spatial policies, relaxing housing supply constraints and subsidizing the relocation of low-income workers, and compare their effects in the model with and without labor

market power.

Each policy is directed at one of two sets of cities. The first comprises New York, San Francisco, and San Jose, the three cities in which Hsieh and Moretti (2019) relax housing supply constraints: large labor markets where many employers compete, with an employment-weighted mean markdown of 6 percent, against 16 percent in the rest of the country. The second comprises the *concentrated cities*, the 31 metropolitan areas whose employment-weighted average Herfindahl–Hirschman index across location-industry cells is at least 0.25, the level above which the 2010 DOJ–FTC Horizontal Merger Guidelines classify a market as highly concentrated (2,500 on their scale, which measures shares in percent; Azar, Marinescu and Steinbaum 2022). An index of 0.25 corresponds to four employers of equal size. These cities are small metropolitan areas, mostly in the Midwest, such as Decatur, Illinois, Sheboygan, Wisconsin, and Muncie, Indiana. At the concentrated cities the mean markdown is 35 percent, against 14 percent in the rest of the country. The concentrated cities account for 3.6 percent of employment, and New York, San Francisco, and San Jose for 11 percent.

We compare the equilibrium with each policy to the benchmark, the no-policy equilibrium of 2019. Both are stationary: the distribution of workers across location-industry cells reproduces itself under the equilibrium choice probabilities. The level of labor income taxes (ς_0) adjusts to balance the government budget and finance the policy, while expenditure on public goods and the progressivity parameter ς_1 stay at their benchmark values.[27] We abstract from transitional dynamics.

We measure the welfare effect of a policy as the percent change in consumption of goods and housing that leaves a worker indifferent between the benchmark and the equilibrium with the policy, averaged over workers with benchmark population weights (Appendix C.1).[28]

We solve each counterfactual in two versions of the model. The first is the model with labor market power, in which wages follow equation (9). The second assumes competitive labor markets, in which firms take wages as given and pay the marginal product of labor, $w^e_{jkt} = \text{MPL}^e_{jkt}$. The model without labor market power attributes low wages in the data to lower productivity A_{jkt}, whereas the model with labor market power attributes part of the same wage variation to markdowns. The two versions share the same migration and switching costs, agglomeration elasticity, and income tax function, and both reproduce the observed 2019 allocation, wages, and rents exactly. The gap between their welfare predictions is therefore the error a researcher would make by evaluating the policy in a model without labor market power calibrated to the same data. Without labor market power, the number of establishments is not determined and plays no role in the equilibrium. To isolate the entry margin, we also solve the model with labor market power holding the number of establishments at its benchmark value.

[27] The housing policy has no direct fiscal cost. The tax level still adjusts under it, because the policy changes wages and moves workers across local labor markets, both of which change the revenue from the labor income tax.

[28] The welfare measure is the welfare of workers. As in Diamond (2016) and Hsieh and Moretti (2019), housing is owned by absentee landlords, so the measure leaves out the change in rental income that a policy causes. With free entry, establishments earn zero profit: operating profits exactly cover the fixed cost of operating, which we interpret as a payment to absentee owners, for example of commercial property, as in Hsieh and Moretti (2019).

Housing Supply The first policy lowers the inverse housing supply elasticity of the targeted cities, the margin that laws such as the ADU legislation described in the introduction aim to relax. The housing supply elasticities come from a regression of the Saiz (2010) inverse elasticities on land unavailability and the Wharton land-use regulation index (Gyourko, Saiz and Summers 2008). At New York, San Francisco, and San Jose, the policy follows the experiment of Hsieh and Moretti (2019), in which the three cities adopt the housing supply regulation of the median U.S. city: we lower the Wharton index of each city to the national median and assign it the inverse elasticity the regression predicts at the reduced index, holding land unavailability and the city's residual fixed. The inverse elasticity falls by 5.5 percent in New York, 6.7 percent in San Francisco, and 1.8 percent in San Jose. At the concentrated cities, we lower the inverse housing supply elasticity of each city by 4.7 percent, the average reduction at New York, San Francisco, and San Jose, so that both sets of cities receive an intervention of the same size. At the concentrated cities the policy is therefore a thought experiment of matched size rather than a reform that follows a regulatory rule.[29]

The housing policy lowers rents and draws workers into the targeted cities. At New York, San Francisco, and San Jose, rents fall by 1.0 percent on average and population rises by 0.4 percent (Figure 3).[30] San Francisco, where regulation falls the most, has the largest response, with rents falling by 1.7 percent and population rising by 0.7 percent. The inflow of workers raises employment and operating profits per establishment, and new establishments enter until profits again cover the fixed cost of operating: the number of establishments rises by 0.25 percent at the three cities. The rest of the country loses workers, and establishments exit there. With more competitors, the markdown at the three cities narrows by 0.012 percentage points, from a level of 6.1 percent, while exit widens it by 0.004 percentage points elsewhere. The labor supply elasticity at the three cities barely moves, falling by 0.03 percent. Lower rents draw workers in and make the workers already there more likely to stay. These two effects nearly offset, and the change in the composition of the workforce lowers the elasticity slightly (Appendix A.4). After-tax wages at the three cities rise by 0.03 percent, both because the markdown narrows and because the marginal product of labor rises as agglomeration raises productivity with city employment (Appendix C.5).

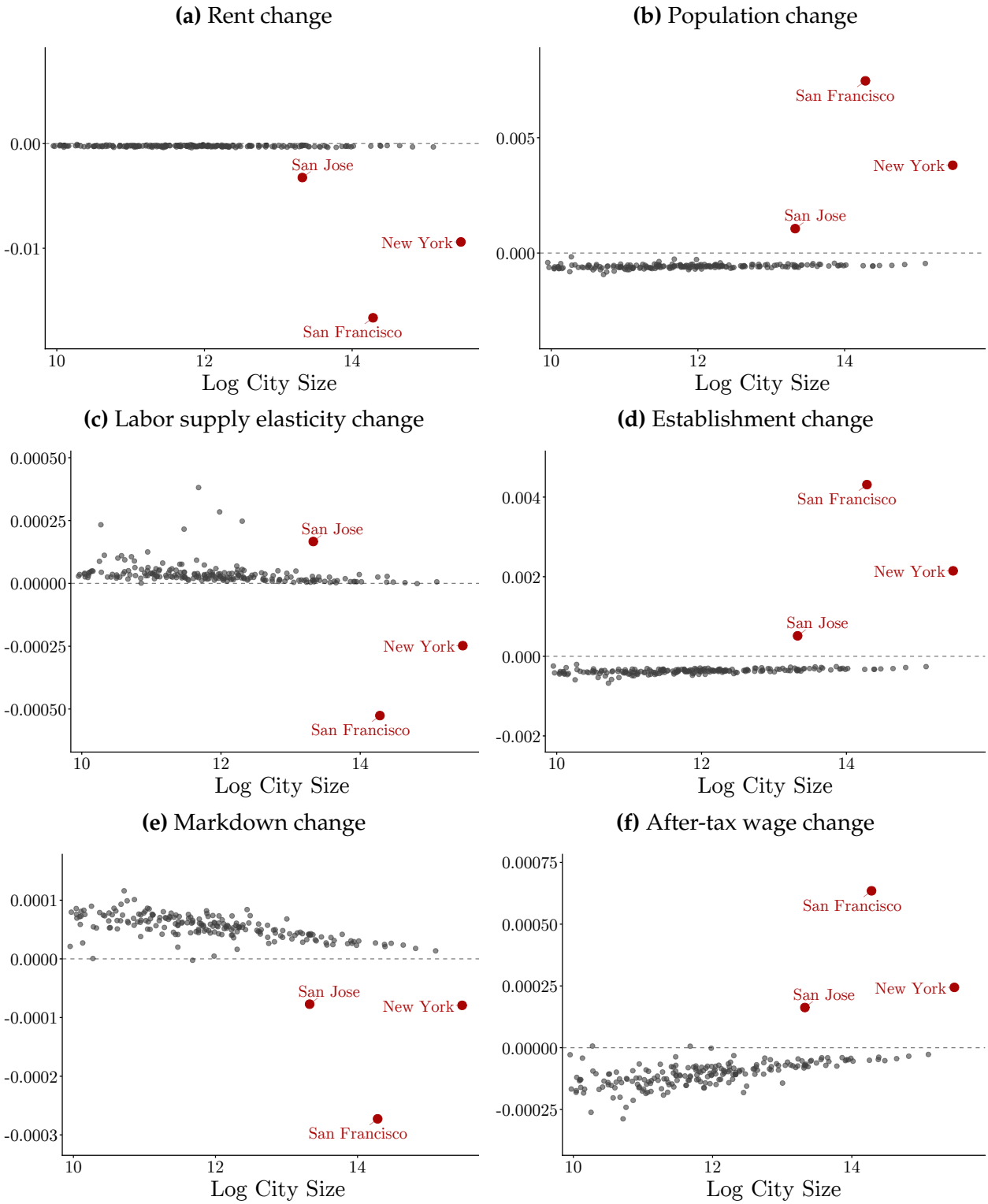
The response at the concentrated cities is larger relative to their size. Rents fall by 1.6 percent and population rises by 1.0 percent (Appendix Figure C3). The number of establishments rises by 0.7 percent, and the markdown narrows by 0.11 percentage points, almost ten times the change at New York, San Francisco, and San Jose. After-tax wages rise by 0.3 per-

[29] Twenty-five of the thirty-one concentrated cities already have a Wharton index at or below the national median, so applying the same rule there would leave their housing supply nearly unchanged.

[30] In the specification of Hsieh and Moretti (2019), the same reform instead raises house prices in the three cities by 10 to 20 percent (Greaney 2026, Table 1). The difference comes from how housing supply is written. Saiz (2010) estimates the inverse elasticity from changes in prices and housing since a base year, so the supply curve should be expressed relative to that base, as in equation (12). Written in levels, as in Hsieh and Moretti (2019), the effect of the reform depends on the arbitrary unit in which population is measured (Greaney 2026). With population measured as a share of the national total, as in their calibration, the population of each city is below one, and a lower inverse elasticity raises the price of housing at any population. With housing measured relative to its 1970 level, the reform lowers house prices in their model by 0.2 to 2.1 percent, the same magnitude as our rent declines.

cent, mostly because of the narrower markdown (Appendix C.5). At both sets of cities, wages move much less than rents.

Figure 3: Housing policy at New York, San Francisco, and San Jose



Notes: Model with labor market power and free entry, main calibration, 2019. Each point is one metropolitan area; targeted cities in red. The horizontal axis is the log of benchmark city population in all panels. Rent is the proportional rent change and population the proportional change in total city population. Panels (c) through (f) are benchmark-employment-weighted city means of cell-level changes across industry and education cells: the log change in the labor supply elasticity, the log change in the number of establishments, the change in the markdown in levels, and the log change in the after-tax wage.

Migration Subsidy The second policy subsidizes moves into the targeted cities, which lowers the cost for workers elsewhere of entering their labor markets. Such policies are standard in the quantitative spatial literature (Giannone et al. 2023); the Moving to Opportunity experiment, which subsidized moves into lower-poverty neighborhoods, is the closest policy precedent (Chetty, Hendren and Katz 2016).

The subsidy pays a fixed transfer of 10 percent of average labor income, about $5,500 in the calibrated 2019 economy, to eligible workers who move into a targeted city; staying in one's own city is never subsidized.[31] Eligibility is income-tested. A worker is eligible if the wage of their cell in the benchmark falls below 60 percent of the employment-weighted median wage, about $27,700, a threshold modeled on the 60 percent income limit of the Low-Income Housing Tax Credit. Under this threshold, 6.4 percent of non-college workers and 0.2 percent of college workers are eligible, or 3.8 percent of total employment, and the eligible share of employment declines with city size (Appendix C.2). Figure 4 shows the cross-section when the subsidy is directed at the concentrated cities, and Appendix Figure C4 when it is directed at New York, San Francisco, and San Jose.

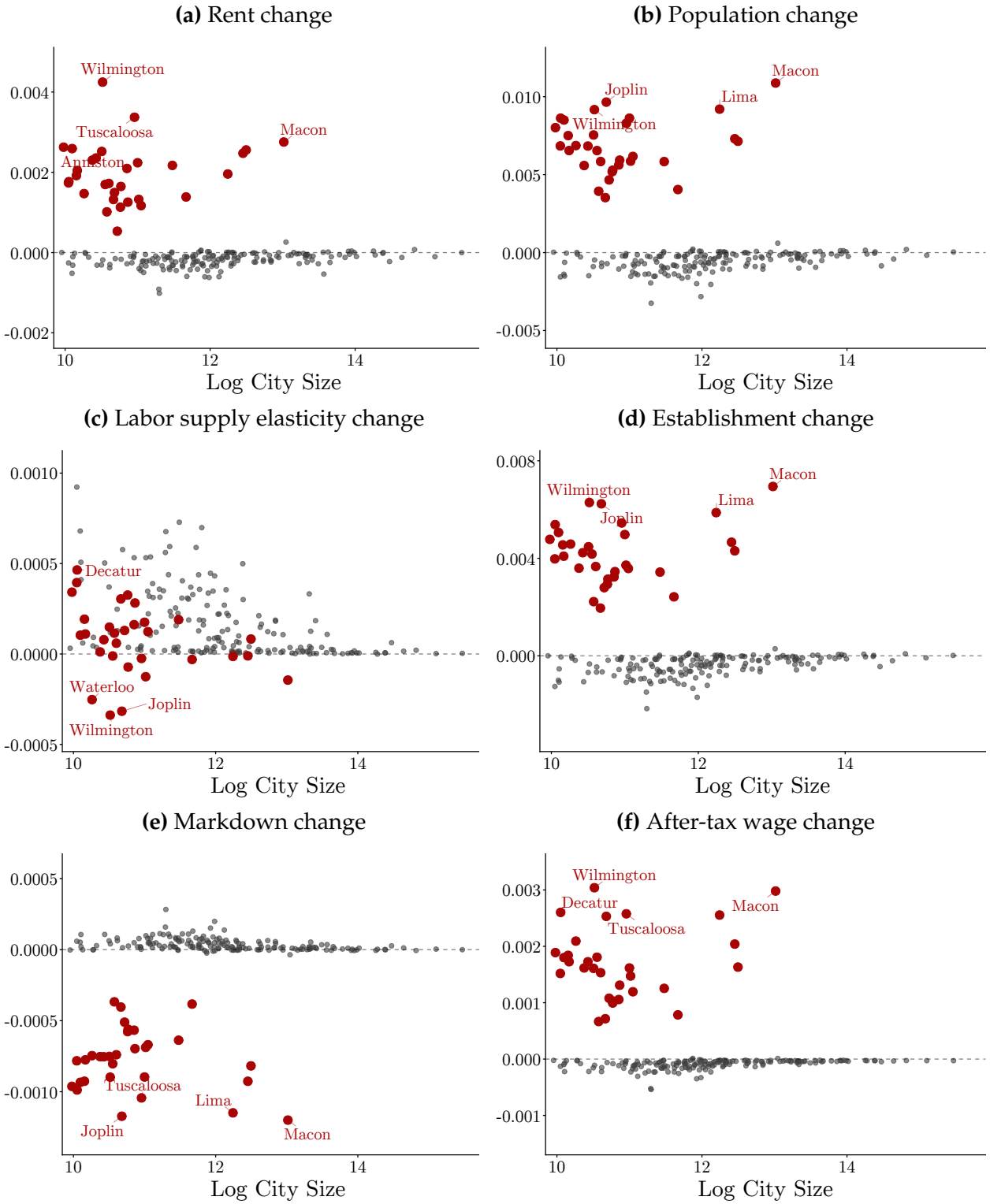
The subsidy lowers the cost of moving into the targeted cities for eligible workers in every other city. Population rises by 0.8 percent at the concentrated cities and by 0.3 percent at New York, San Francisco, and San Jose, where the arrivals are small relative to the existing stock. The arrivals raise housing demand and push rents up, by 0.2 and 0.1 percent. Two effects on the labor supply elasticity at the targeted cities nearly offset. The workers the subsidy draws in would not have moved without it and are less attached to their new city than the workers already there, which raises the elasticity. As the cities grow, however, a larger share of the workers choosing among markets already work there, and these workers stay with a probability close to one, which lowers the elasticity. On net, the labor supply elasticity at the targeted cities barely changes (Appendix A.4). At the concentrated cities, the arrivals also raise operating profits per establishment, and the number of establishments rises by 0.5 percent. With more competitors, the markdown narrows by 0.09 percentage points and after-tax wages rise by about 0.2 percent. At New York, San Francisco, and San Jose, where markdowns are already low, the markdown narrows by 0.005 percentage points and after-tax wages barely move.

The two policies change markdowns through different mechanisms. Both draw workers into the targeted cities, and the arrivals attract establishments whose entry narrows markdowns there. The housing policy also lowers rents for the workers already in these cities, who become more likely to stay, and the labor supply elasticity there falls. The subsidy instead lowers the cost of moving for eligible workers everywhere else, who become less likely to stay in the markets where they work, and the labor supply elasticity in these markets rises. Appendix A.4 decomposes the change in the elasticity under each policy into the response of incumbents, of outsiders, and of the composition of the workforce.

Welfare and the Role of Labor Market Power With labor market power, the welfare gain

[31] The transfer is comparable in scale to the moving assistance provided by the Moving to Opportunity experiment, roughly $5,700 per family in 2019 dollars (Chetty, Hendren and Katz 2016).

Figure 4: Migration subsidy directed at the concentrated cities



Notes: Model with labor market power and free entry, main calibration, 2019. Each point is one metropolitan area; targeted cities in red. The targeted cities are the concentrated cities, the 31 metropolitan areas whose employment-weighted average Herfindahl–Hirschman index is at least 0.25. The horizontal axis is the log of benchmark city population in all panels. Rent is the proportional rent change and population the proportional change in total city population. Panels (c) through (f) are benchmark-employment-weighted city means of cell-level changes across industry and education cells: the log change in the labor supply elasticity, the log change in the number of establishments, the change in the markdown in levels, and the log change in the after-tax wage.

of both policies is smaller than a model without it predicts when they target New York, San Francisco, and San Jose, and larger when they target the concentrated cities (Table 4). The housing policy raises welfare by 0.039 percent of consumption when it targets New York, San Francisco, and San Jose, and by 0.029 percent when it targets the concentrated cities. These gains are 8.3 percent smaller and 32.6 percent larger than a model without labor market power predicts. The percentage differences are nearly the same for both policies when the reduction in the inverse housing supply elasticity and the transfer per mover are doubled.

Table 4: Welfare effects of the housing policy and the migration subsidy

	Labor market power			Difference	
	Without	With			
		Entry fixed	Free entry	Points	Percent
Housing policy, 1×: inverse supply elasticity 4.7 percent lower					
New York, San Francisco, San Jose					
1×	0.0425	0.0427	0.0390	−0.0035	−8.3
2×	0.0887	0.0890	0.0813	−0.0074	−8.3
Concentrated cities					
1×	0.0216	0.0212	0.0286	0.0070	32.6
2×	0.0439	0.0431	0.0583	0.0144	32.8
Income-tested migration subsidy, 1×: $5,500 per mover					
New York, San Francisco, San Jose					
1× Eligible workers	0.0421	0.0456	0.0374	−0.0047	−11.1
Whole population	−0.0007	−0.0005	−0.0028	−0.0022	–
2× Eligible workers	0.0825	0.0894	0.0733	−0.0092	−11.1
Whole population	−0.0015	−0.0011	−0.0057	−0.0042	–
Concentrated cities					
1× Eligible workers	0.1062	0.1195	0.1195	0.0133	12.6
Whole population	0.0016	0.0024	0.0064	0.0048	–
2× Eligible workers	0.2078	0.2339	0.2338	0.0260	12.5
Whole population	0.0025	0.0042	0.0118	0.0093	–

Notes: Welfare gains in percent of consumption of goods and housing, averaged over the group shown with benchmark population weights; main calibration, 2019. Eligible workers are those whose benchmark cell wage is below 60 percent of the employment-weighted median wage (3.8 percent of employment). Entry fixed holds the number of establishments at its benchmark value. The difference is the free-entry column less the column without labor market power, in percentage points and as a percent of the latter. The percent is omitted for the whole population under the subsidy, where the gain without labor market power is close to zero.

For the housing policy, the difference between the two models comes from establishment entry. With the number of establishments held fixed, the gain with labor market power is within 2 percent of the gain without it whichever cities the policy targets. Under free entry, establishments enter the targeted cities as workers arrive and exit the rest of the country as workers leave, so markdowns narrow in the targeted cities and widen elsewhere. By equation (9), a given proportional increase in the number of establishments narrows markdowns more where they are high.[32] The net effect therefore depends on the markdown at the targeted cities relative to the rest of the country. Holding employment at its benchmark distribution,

[32]The markdown in equation (9) is $\psi = 1/(1+x)$, with $x = m\,\eta\,(1-\varsigma_1)$. Holding the labor supply elasticity

the national mean markdown, 14.5 percent in 2019, falls by 0.002 percentage points when the policy targets the concentrated cities, where the markdown is 35 percent, and rises by 0.002 percentage points when it targets New York, San Francisco, and San Jose, where it is 6 percent.[33] The gain with labor market power is correspondingly larger than without it in the first case and smaller in the second.

The subsidy raises the welfare of eligible workers by 0.037 percent of consumption when it targets New York, San Francisco, and San Jose, and by 0.120 percent when it targets the concentrated cities. These gains are 11.1 percent smaller and 12.6 percent larger than a model without labor market power predicts. By lowering the cost of moving to the targeted cities, the subsidy makes eligible workers less likely to stay in the markets where they work, which raises the labor supply elasticity and narrows the markdown in these markets (Appendix A.4). With the number of establishments held fixed, the gains are therefore 8.3 and 12.6 percent larger than without labor market power. Under free entry, the workers the subsidy draws away leave these same markets, which lose more establishments than other markets (Appendix C.4). When the subsidy targets New York, San Francisco, and San Jose, the effect of this exit outweighs that of the more elastic labor supply. When it targets the concentrated cities, eligible workers who live there gain from establishment entry about as much as those elsewhere lose from exit, and the difference between the two models is the same as with the number of establishments held fixed.

For the whole population, the welfare effect of the subsidy is close to zero, because the transfer is financed by all taxpayers and received only by the eligible workers who move. It is positive in both models when the subsidy targets the concentrated cities. When it targets New York, San Francisco, and San Jose, the model with labor market power and free entry gives a loss of 0.003 percent of consumption, while the model without labor market power and the model with the number of establishments held fixed give smaller losses, of 0.0007 and 0.0005 percent. The loss reflects the same entry at the targeted cities and exit elsewhere as under the housing policy: under free entry the subsidy raises the national mean markdown, with employment held at its benchmark distribution, by 0.001 percentage points.

Two further sets of targeted cities, with markdowns between those of New York, San Francisco, and San Jose and those of the concentrated cities, give differences between the two models that lie in between (Appendix Table C3). Each set holds about the same share of employment as the concentrated cities and is taken from the ranking of cities by employer concentration. In the first, Baltimore, Providence, Portland, and West Palm Beach, the mean

fixed, $\partial\psi/\partial \log m = -x/(1+x)^2 = -\psi\,(1-\psi)$. The product $\psi\,(1-\psi)$ rises with ψ as long as $\psi < 1/2$, which holds for the averages we consider: 6 percent at New York, San Francisco, and San Jose, 35 percent at the concentrated cities, and 14.5 percent nationally.

[33] Letting employment move with the policy, the national mean markdown rises by 0.007 percentage points when the policy targets the concentrated cities, where markdowns are high, and falls by 0.004 percentage points when it targets New York, San Francisco, and San Jose, where they are low, because workers move toward the targeted cities. This change in the composition of employment adds little to workers' welfare, because the workers who move were close to indifferent between locations before the policy. Workers who would work in a market with or without the policy instead bear the full change in its markdown. The change reported in the text averages these within-market changes with benchmark employment weights, and its sign matches the sign of the difference between the welfare gains with and without labor market power.

markdown is 11 percent, against 15 percent in the rest of the country, and the gain with labor market power is 3.8 percent smaller than a model without it predicts for the housing policy and 5.8 percent smaller for eligible workers under the subsidy. In the second, eleven medium-sized cities such as Albuquerque, Louisville, and Richmond, the mean markdown is 21 percent, against 14 percent, and the gains are 7.3 and 5.4 percent larger. Across the four sets, the difference between the two models rises with the gap between the markdown in the targeted cities and that in the rest of the country (Appendix Figure C2). Only when the markdown in the targeted cities is close to that in other cities do the two models give similar welfare gains for both policies.

Omitting labor market power changes the predicted welfare gains of the two policies more than omitting agglomeration economies, which quantitative spatial models commonly include. In the model with labor market power, setting the agglomeration elasticity to zero rather than 0.04 (Melo, Graham and Noland 2009) changes the gain from the housing policy by 0.9 percent when it targets New York, San Francisco, and San Jose and by 3.0 percent when it targets the concentrated cities, and the gain of eligible workers from the subsidy by 5.1 and 0.4 percent. The differences between the models with and without labor market power are 8 to 33 percent. Appendix C.3 reports how the welfare gain of each policy changes when each general-equilibrium channel is shut off in turn: agglomeration, the tax adjustment that balances the government budget, pre-tax wages, rents, and establishment entry.

Shift-Share Calibration Under the shift-share calibration, with the number of establishments held fixed, the gain of eligible workers from the subsidy is 6.9 and 12.4 percent larger with labor market power than without it, against 8.3 and 12.6 percent under the main calibration, and for the housing policy the two models differ by at most 3 percent (Appendix D).

7 Conclusion

We develop a spatial equilibrium model in which wage markdowns arise from migration costs, industry-switching costs, and imperfect competition among firms, and calibrate it to four decades of U.S. data and to the firm-level labor supply elasticity estimated in the monopsony literature.

Local labor market power has declined since 1980: the average wage markdown roughly halved despite declining spatial mobility, mostly because the number of local establishments grew and industry-switching costs fell. The decomposition is similar when the wage coefficient is set to its shift-share estimate and when prices, populations, and establishment counts adjust in general equilibrium.

The welfare gains of the two spatial policies we study differ by 8 to 33 percent between the models with and without labor market power. Both policies draw workers from the rest of the country to the targeted cities, so that establishments enter these cities and exit the rest of the country. Markdowns respond more to establishment entry and exit where they are

high. A model without labor market power therefore overstates the welfare gain when the markdown in the targeted cities is lower than in the rest of the country, as in large productive cities, and understates it when the markdown is higher, as in concentrated labor markets, by more the larger the difference. The migration subsidy also makes the labor supply of eligible workers more elastic, which narrows markdowns in the markets where they work and partly offsets the effect of establishment exit. The two models give similar welfare gains only when the markdown in the targeted cities is close to that in the rest of the country.

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Appendix

This Appendix is organized as follows. Section A provides details about the model. Section B presents further details on the data and the estimation. Section C provides additional information on the policy counterfactuals. Section D reports the shift-share estimate of the wage coefficient and repeats the counterfactuals under the shift-share calibration. Section E reports the general-equilibrium variant of the markdown decomposition. Finally, Section F contains additional tables and figures referenced in the text.

A Model

A.1 Labor Supply Derivative

The labor supply curve for labor of educational type e is given by aggregating the choice probabilities across all individuals i of type e.

$$\begin{aligned} N^e_{j'k't} &= \sum_{i=1}^{N^e} p^e_{ij'k't}(j,k) \\ &= \sum_{i=1}^{N^e} \frac{\exp\left(\beta_w \log(y^e_{j'k't}) + \beta_r \log(r_{j't}) + \xi^e_{j'k't} - \mu^e_t(j(i),j') - \delta^e_t(k(i),k')\right)}{\sum_{j'',k''} \exp\left(\beta_w \log(y^e_{j''k''t}) + \beta_r \log(r_{j''t}) + \xi^e_{j''k''t} - \mu^e_t(j(i),j'') - \delta^e_t(k(i),k'')\right)} \end{aligned} \tag{15}$$

Denote as a shorthand

$$\begin{aligned} f^e_{ij'k't} &\equiv \exp\left(\beta_w \log(y^e_{j'k't}) + \beta_r \log(r_{j't}) + \xi^e_{j'k't} - \mu^e_t(j(i),j') - \delta^e_t(k(i),k')\right) \\ \Rightarrow N^e_{j'k't} &= \sum_{i=1}^{N^e} \frac{f^e_{ij'k't}}{\sum_{j''k''} f^e_{ij''k''t}} \end{aligned}$$

Consider the labor supply $N^e_{j'k't}$ in location-industry $j'k'$ as a function of its own after-tax wage $y^e_{j'k't}$. The own after-tax wage $y^e_{j'k't}$ appears in the numerator and denominator of every individual-level summand of equation (15). Within each individual-level summand, $y^e_{j'k't}$ is part of only one summand in the denominator. Therefore, the derivative of the denominator with respect to $y^e_{j'k't}$ is the same as the derivative of the numerator.

$$\begin{aligned} \frac{\partial N^e_{j'k't}}{\partial y^e_{j'k't}} &= \sum_{i=1}^{N^e} \frac{f^e_{ij'k't}\frac{\beta_w}{y^e_{j'k't}}(\sum_{j''k''} f^e_{ij''k''t}) - f^e_{ij'k't}f^e_{ij'k't}\frac{\beta_w}{y^e_{j'k't}}}{(\sum_{j''k''} f^e_{ij''k''t})^2} \\ &= \frac{\beta_w}{y^e_{j'k't}}\left(\sum_{i=1}^{N^e} \frac{f^e_{ij'k't}}{(\sum_{j''k''} f^e_{ij''k''t})} - \sum_{i=1}^{N^e} \frac{f^e_{ij'k't}f^e_{ij'k't}}{(\sum_{j''k''} f^e_{ij''k''t})^2}\right) \\ &= \frac{\beta_w}{y^e_{j'k't}}\left(\sum_{i=1}^{N^e} p^e_{ij'k't} - \sum_{i=1}^{N^e} \left(p^e_{ij'k't}\right)^2\right) \end{aligned}$$

or

$$\frac{\partial N^e_{j'k't}}{\partial y^e_{j'k't}} = \frac{\beta_w}{y^e_{j'k't}} \left(\sum_{i=1}^{N^e} p^e_{ij'k't}(1 - p^e_{ij'k't}) \right) \tag{16}$$

Tax adjustment The derivative above is taken with respect to the wage that enters workers' utility, the after-tax wage y. Firms set the pre-tax wage w. Since $y = w\,(1 - T_t(w))$ with $1 - T_t(w) = \varsigma_0 (w/\overline{w}_t)^{-\varsigma_1}$, differentiation gives $\partial \log y / \partial \log w = 1 - \varsigma_1$. By the chain rule, the elasticity of labor supply with respect to the pre-tax wage equals the elasticity in equation (6) multiplied by $(1 - \varsigma_1)$, the factor that appears in the wage-setting condition (9).

A.2 Labor Supply Derivative Including Impacts on Local Rents

If we assume that firms take into account that after-tax wages $y^e_{j'k't}$ may affect local rents $r_{j't}$, we obtain the following expression for the labor supply derivative as considered by strategic firms:

$$\frac{\partial N^e_{j'k't}}{\partial y^e_{j'k't}} = \left(\frac{\beta_w}{y^e_{j'k't}} + \frac{\beta_r}{r_{j't}} \cdot \frac{\partial r_{j't}}{\partial y^e_{j'k't}} \right) \left(\sum_{i=1}^{N^e} p^e_{ij'k't}(1 - p^e_{ij'k't}) \right) \tag{17}$$

Counterfactual exercises that use equation (17) instead of the expression in (6) show that incorporating the effect of wages on rents moves the implied median labor supply elasticity by less than 0.01.

A.3 Equilibrium

An equilibrium is a distribution of workers $(N^n_{j'k't}, N^c_{j'k't})$ across location-industries, pre-tax wages $(w^n_{j'k't}, w^c_{j'k't})$, with after-tax wages $y^e_{j'k't} = (1 - T_t(w^e_{j'k't}))\, w^e_{j'k't}$, a number of establishments $m_{j'k't}$ in each location-industry, rents $r_{j't}$, and labor income tax level ς_0 such that

1. Local labor markets clear for both types of labor,

$$\begin{aligned} N^n_{j'k't} = m_{j'k't} n^n &= \sum_i p^n_{ij'k't} \\ N^c_{j'k't} = m_{j'k't} n^c &= \sum_i p^c_{ij'k't} \end{aligned}$$

2. Firms' labor demand $(n^n_{j'k't}, n^c_{j'k't})$ is an equilibrium of the augmented Cournot game within each local labor market, where each firm chooses demand for each type of labor to maximize their profits, taking as given the demand of all other firms and labor supply elasticities:

$$A_{j'k't} \left(\theta^c_{j'k't} (n^c)^{\frac{\rho-1}{\rho}} + \theta^n_{j'k't} (n^n)^{\frac{\rho-1}{\rho}} \right)^{\frac{1}{\rho-1}} \theta^e_{j'k't} \left(n^e\right)^{-\frac{1}{\rho}} - \left. \frac{\partial w^e_{j'k't}}{\partial n^e} \right|_{N^e_{j'k't}} n^e - w^e_{j'k't} = 0$$

In the counterfactuals, productivity $A_{j'k't}$ adjusts with city employment through equa-

tion (10). Firms satisfy the free entry condition $\pi^*_{j'k't} = F_{j'k't}$; in the variant with the number of establishments held fixed, this condition is replaced by $m_{j'k't}$ equal to its benchmark value.

3. Housing market supply $h^S_{j't}$ equals demand $h^D_{j't}$ in each location j' and year t:

$$\begin{aligned} h^S_{j't} &= h_{j',1980}\left(\frac{r_{j't}}{\kappa_{j't}}\right)^{\frac{1}{\gamma_{j't}}} \\ &= \sum_{k''} N^c_{j'k''t}\frac{(-\beta_r)y^c_{j'k''t}}{\beta_w r_{j't}} + \sum_{k''} N^n_{j'k''t}\frac{(-\beta_r)y^n_{j'k''t}}{\beta_w r_{j't}} \\ &= h^D_{j't} \end{aligned}$$

4. Workers' choices of locations and industries solve their utility maximization problems described in equation (1).

5. The level of taxes ς_0 balances the Government budget, i.e.,

$$\mathcal{T}_t = \overline{G}_t + G^p_t,$$

where tax revenues are given by

$$\mathcal{T}_t = \sum_{j''}\sum_{k''} N^c_{j''k''t} w^c_{j''k''t}\left(1-\varsigma_0\left(\frac{w^c_{j''k''t}}{\overline{w}_t}\right)^{-\varsigma_1}\right) + \sum_{j''}\sum_{k''} N^n_{j''k''t} w^n_{j''k''t}\left(1-\varsigma_0\left(\frac{w^n_{j''k''t}}{\overline{w}_t}\right)^{-\varsigma_1}\right),$$

$\overline{G}_t$ denotes expenditure on public goods, and G^p_t denotes expenditure on policies.

A.4 How Policies Change the Labor Supply Elasticity

Each worker contributes $p^e_{ij'k't}(1-p^e_{ij'k't})$ to the slope of labor supply in equation (6), a term that is largest for workers indifferent between the market and their best alternative, $p = 1/2$. Divided by employment, the same terms give $\eta = \beta_w(1-\bar{p})$, which falls in the employment-weighted average choice probability $\bar{p}$ (footnote to equation (6)). Dropping education and time indices and grouping workers by their origin cell o, the average choice probability of a destination cell d is

$$\bar{p}_d(N, v) = \frac{\sum_o N_o\, p_{od}(v)^2}{\sum_o N_o\, p_{od}(v)}, \tag{18}$$

where N_o is the number of workers in origin o and $p_{od}(v)$ is the probability that a worker in o chooses d, given the mean utilities v of all cells, which include after-tax wages, rents, and any subsidy to moving. A policy moves the stationary equilibrium from (N^0, v^0) to (N^1, v^1), and we decompose the change as

$$\Delta\bar{p}_d = \underbrace{\bar{p}_d(N^0, v^1) - \bar{p}_d(N^0, v^0)}_{\text{attachment}} + \underbrace{\bar{p}_d(N^1, v^0) - \bar{p}_d(N^0, v^0)}_{\text{reallocation}} + \underbrace{R_d}_{\text{interaction}}. \tag{19}$$

Attachment holds the population of every origin at its benchmark value and moves the choice probabilities to their values with the policy. Reallocation holds the choice probabilities at their benchmark values and moves the populations. The interaction R_d is the remainder, which is not zero because $\bar{p}_d$ is a ratio of two sums. Attachment splits exactly by origin. With $m_1 = \sum_o N_o p_{od}$ and $m_2 = \sum_o N_o p_{od}^2$, let $\Delta m_{1,g}$ and $\Delta m_{2,g}$ be the change in the terms of origin group g between (N^0, v^0) and (N^0, v^1). The contributions

$$C_g = \frac{\Delta m_{2,g} - \bar{p}_d(N^0, v^1)\,\Delta m_{1,g}}{m_1(N^0, v^0)}$$

sum over groups to the attachment term. Incumbents are the workers whose origin is the cell itself, $o = d$, and outsiders all other origins, in the same city or elsewhere. Since $\eta_d = \beta_w(1 - \bar{p}_d)$, a change $\Delta\bar{p}_d$ changes the elasticity by $-\Delta\bar{p}_d/(1 - \bar{p}_d)$ in proportion, and Table A1 reports each term in these units.

Reallocation always moves the elasticity in the opposite direction to the population of the cell, as the table shows in every row. In the stationary equilibrium, the workers of a cell are also its origin population. When a cell gains workers, a larger share of its workforce starts in the cell itself and chooses it again with a probability close to one, which raises $\bar{p}_d$ and lowers the elasticity. When it loses workers, the opposite happens.

The housing policy lowers rents in the targeted cities and so raises the value of living there only. There it makes incumbents more likely to stay, which raises $\bar{p}$, and outsiders more likely to arrive, which lowers it. At the concentrated cities the two changes nearly offset, moving the elasticity by -0.073 and 0.066 percent. The cities gain workers, so reallocation lowers the elasticity by 0.108 percent, and the elasticity falls.

The migration subsidy works through outsiders. It lowers the cost of moving into the targeted cities for eligible workers in every origin. Each eligible worker employed elsewhere then chooses a targeted market with a probability that remains small, which lowers $\bar{p}$ there, while the probability that the worker stays in their current market falls, which raises the elasticity in the markets they might leave. Incumbents become slightly less likely to stay: the subsidy rewards only moves and, unlike under the housing policy, rents in the targeted cities rise in general equilibrium as the arrivals increase housing demand. At the concentrated cities the outsider term raises the elasticity by 0.078 percent, against 0.002 percent for incumbents. The cities also gain workers, and reallocation lowers the elasticity by 0.077 percent, less than the outsider term, so the elasticity rises. Outside the targeted cities the elasticity rises slightly under both policies.

Table A1: Sources of the change in the labor supply elasticity

	Labor supply elasticity					
	Total	Incumbents	Outsiders	Reallocation	Interaction	Population
Targeted cities						
Housing policy						
New York, San Francisco, San Jose	−0.0273	−0.0160	0.0152	−0.0263	−0.0001	0.436
Concentrated cities	−0.1149	−0.0726	0.0660	−0.1075	−0.0009	1.005
Migration subsidy						
New York, San Francisco, San Jose	0.0041	0.0024	0.0115	−0.0097	−0.0001	0.252
Concentrated cities	0.0022	0.0021	0.0784	−0.0774	−0.0010	0.759
Other cities						
Housing policy						
New York, San Francisco, San Jose	0.0021	0.0013	−0.0013	0.0021	0.0000	−0.054
Concentrated cities	0.0053	0.0032	−0.0029	0.0049	0.0000	−0.037
Migration subsidy						
New York, San Francisco, San Jose	0.0017	0.0011	−0.0002	0.0008	0.0000	−0.032
Concentrated cities	0.0062	0.0039	−0.0013	0.0036	0.0000	−0.028

Notes: Percent changes, averaged over location-industry cells and the two education types with benchmark population weights; main calibration, 2019, free entry. Rows name the cities the policy targets; the top panel reports those cities and the bottom panel all other cities. The elasticity columns are the terms of equation (19), with attachment split into incumbents and outsiders, and sum to the total up to rounding. Population is the number of workers in the cells.

B Estimation

B.1 Data

Locations Metropolitan areas are observed in the Census data for individuals who reside in PUMAs that are entirely contained in delineated metro areas. Most metro areas are fully and correctly identified as a collection of PUMAs. The Census data does not identify the metro area for PUMAs that straddle metro areas, located typically on the outskirts. In this case, we do not consider the residents to be residents of the metro area.[34]

Migration and Industry-Switching Flows The ACS samples used for 2011 and 2019 record only the location of residence one year prior, so for those years we construct five-year migration rates by applying state-specific ratios of five-year to one-year migration computed from the 2000 Census, which provides both horizons, assuming this relationship remains stable within states over time.

For industry switching, no survey of comparable size records the industry held five years earlier, so we convert the one-year CPS flow tables to a five-year equivalent using the Panel Study of Income Dynamics: for each 1-digit industry pair and education group we compute the ratio of the five-year to the one-year transition probability, pooled across the 1968–2021 waves, and rescale the one-year tables cell by cell. The five-year window lowers the share of

[34]The percentage of the population by metro area that is misidentified can be found here: https://usa.ipums.org/usa/volii/incompmetareas.shtml. Metro areas in which a nontrivial share of residents are misidentified as non-residents are mostly Puerto Rican metro areas, which we do not consider in this study, and a few metro areas in geographically small east coast states, e.g. Worcester MA, Waterbury CT, New-Haven Meriden CT, and Nashua NH.

workers who remain in their industry by roughly 7.5 percentage points on average, with no discernible trend across decades, so a single pooled ratio is applied to every year.

Market Level Wages To measure location-industry specific wages $\log(w_{jkt})$, we regress individual annual pre-tax wage and salary income, deflated to 2010 dollars, on demographics and location-by-industry fixed effects, separately for each educational group and year t:

$$\log(w^e_{ijkt}) = \beta^e_{1t}\text{age}_{it} + \beta^e_{2t}\text{gender}_{it} + \beta^e_{3t}\text{race}_{it} + \alpha_{j(i)k(i)} + \varepsilon_{it} \tag{20}$$

We measure age in bins (below 30, 30-40, 40-50, 50-60, 60+) to account for non-linearities in wages by age. Therefore, in a slight abuse of notation, age_{it} stands for a set of count variables and β_{1t} stands for a set of four coefficients, with 30-40 being the omitted reference group. We define the market-level wage index $\log(w_{jkt})$ for each period t as the location-by-industry fixed effect α_{jk} obtained from this regression.

Location and Industry-Level Wages The estimation of location-specific wages, $\log(w_{jt})$, follows the same approach as in equation (20), with the addition of industry fixed effects to account for differences in industry composition across locations:

$$\log(w^e_{ijt}) = \beta^e_{1t}\text{age}_{it} + \beta^e_{2t}\text{gender}_{it} + \beta^e_{3t}\text{race}_{it} + \alpha_{j(i)} + \alpha_{k(i)} + \varepsilon_{it} \tag{21}$$

In this specification, $\alpha_{j(i)}$ captures the location-level wage component, controlling for observable worker characteristics and industry composition. Similarly, $\alpha_{k(i)}$ captures the industry-level wage component, obtained while controlling for location fixed effects.

Housing Rents Analogously to wages, we calculate a rental index for each location j and year t in the sample to facilitate comparisons of the annual cost of housing across locations. We follow Bayer, Ferreira and McMillan (2007) and Bishop et al. (2024) by pooling data on rents and property values. As the dependent variable, we use annual gross rent for renters and property value for home owners. We estimate separately for each year:

$$\log(p_{ijt}) = \beta_{1t}\mathbf{x}_{it} + \beta_{2t}\text{own}_{it} + \sum_s \beta_{3st}\ \text{own}_{it} \cdot I_{st} + \alpha_{j(i)t} + \varepsilon_{it} \tag{22}$$

$\mathbf{x}_{it}$ is a vector of housing characteristics including the number of bedrooms, the age of the structure, and the type of unit. The indicator own_{it} controls for owner-occupancy. We allow the ratio of home values to annual gross rents to vary by state by adding a set of state indicator variables I_{st} interacted with the ownership indicator. We define the market-level rent index $\log(r_{jt})$ as the location fixed effect α_{jt}.

B.2 Housing and Labor Markets

Housing Market To estimate the remaining housing supply parameters, we substitute the model-based expressions for housing demand (equation 11) into the inverse housing supply

functions (equation 12) and solve for $\log(r_{jt})$:

$$\begin{aligned}
\log(r_{jt}) &= \log(\kappa_{jt}) + \gamma_{jt} \log\left(\sum_k N^n_{jkt} \frac{-\beta_r y^n_{jkt}}{\beta_w r_{jt}} + \sum_k N^c_{jkt} \frac{-\beta_r y^c_{jkt}}{\beta_w r_{jt}}\right) \\
(1+\gamma_{jt})\log(r_{jt}) &= \log(\kappa_{jt}) + \gamma_{jt} \log\left(\sum_k N^n_{jkt} \frac{-\beta_r y^n_{jkt}}{\beta_w} + \sum_k N^c_{jkt} \frac{-\beta_r y^c_{jkt}}{\beta_w}\right) \\
\log(r_{jt}) &= \underbrace{\frac{\log(\kappa_{jt})}{(1+\gamma_{jt})}}_{\equiv \tilde{\kappa}_{jt}} + \frac{\gamma_{jt}}{(1+\gamma_{jt})} \log\left(\sum_k N^n_{jkt} \frac{-\beta_r y^n_{jkt}}{\beta_w} + \sum_k N^c_{jkt} \frac{-\beta_r y^c_{jkt}}{\beta_w}\right) \qquad (23)
\end{aligned}$$

Using the market-level rents and wages (equations 20 and 22), with after-tax wages obtained from the tax function of Section 3, inverse housing supply elasticities γ_{jt} from Saiz (2010), and the values of β_w and β_r described in Section 4.3, we then choose $\tilde{\kappa}_{jt}$ to equate the left and right-hand sides of equation (23). The derivation writes housing supply without the normalization by $h_{j,1980}$ of equation (12); the normalization is absorbed into κ_{jt} and leaves the calibrated equilibrium unchanged.

Labor Market The parameters of interest on the labor demand side are total factor productivity A_{jkt} and factor shares θ^c_{jkt} and θ^n_{jkt}, the parameters of the production function. For each year and location-industry, we estimate these using the strategic first-order conditions:

$$\begin{aligned}
\frac{\partial\, \pi_{jkt}}{\partial\, n^n} &= A_{jkt}\left(\theta^c_{jkt}(n^c)^{\frac{\rho-1}{\rho}} + \theta^n_{jkt}(n^n)^{\frac{\rho-1}{\rho}}\right)^{\frac{1}{\rho-1}} \theta^n_{jkt}\,(n^n)^{-\frac{1}{\rho}} - \left.\frac{\partial w^n_{jkt}}{\partial n^n}\right|_{N^n_{jkt}} n^n - w^n_{jkt} \\
&\stackrel{!}{=} 0 \qquad (24) \\
\frac{\partial\, \pi_{jkt}}{\partial\, n^c} &= A_{jkt}\left(\theta^c_{jkt}(n^c)^{\frac{\rho-1}{\rho}} + \theta^n_{jkt}(n^n)^{\frac{\rho-1}{\rho}}\right)^{\frac{1}{\rho-1}} \theta^c_{jkt}\,(n^c)^{-\frac{1}{\rho}} - \left.\frac{\partial w^c_{jkt}}{\partial n^c}\right|_{N^c_{jkt}} n^c - w^c_{jkt} \\
&\stackrel{!}{=} 0 \qquad (25)
\end{aligned}$$

Combining the two conditions and rearranging yields

$$\frac{\theta^n_{jkt}}{\theta^c_{jkt}} = \frac{\left.\frac{\partial w^n_{jkt}}{\partial n^n}\right|_{N^n_{jkt}} n^n + w^n_{jkt}}{\left.\frac{\partial w^c_{jkt}}{\partial n^c}\right|_{N^c_{jkt}} n^c + w^c_{jkt}} \left(\frac{n^n}{n^c}\right)^{\frac{1}{\rho}} \qquad (26)$$

Location-industry specific wages and employment are taken from the data. We obtain the slopes of the local labor supply curves $\left.\frac{\partial w^n_{jkt}}{\partial n^n}\right|_{N^n_{jkt}}$ and $\left.\frac{\partial w^c_{jkt}}{\partial n^c}\right|_{N^c_{jkt}}$ directly from the derivative of the labor supply functions in equation (5) with respect to the after-tax wage, converted to the pre-tax wage through the tax function, using the inverse function theorem.[35] Following Card (2009), we set ρ to 2. Using the assumption $\theta^c_{jkt} + \theta^n_{jkt} = 1$, we now have estimates for factor shares θ^c_{jkt} and θ^n_{jkt} and can plug these into either equation (24) or (25) to recover A_{jkt}. In the

[35] See Appendix Section A.1 for derivations.

counterfactual exercises, we keep factor shares and the exogenous component of productivity, a_{jkt} in equation (10), fixed; productivity A_{jkt} moves only through agglomeration.

Wage markdowns With productivity and factor share estimates in hand, we calculate counterfactual competitive wages $\hat{w}_{jkt}^{comp,e}$, the wages that would emerge if firms behaved as price-takers: setting the labor supply derivative terms in equations (24) and (25) to zero equates each wage to the marginal product of labor, the term MPL_{jkt}^{e} in equation (9). The resulting markdowns depend on the number of symmetric Cournot players in each market and on the inverse labor supply elasticities in each location-industry, which come from our structural model of labor supply. We calculate markdowns as the proportional gap between observed and competitive wages:

$$\psi_{jkt}^{e} = 1 - \frac{w_{jkt}^{e}}{\hat{w}_{jkt}^{comp,e}} \tag{27}$$

B.3 First-Step Estimation

This section details the estimation of the moving and switching costs described in Section 4.2. The utility cost of moving varies across educational types e and over time. Equation (28) parameterizes it as a fixed cost paid for any move plus a piecewise constant function of physical distance, which accommodates non-linearities in distance. The first term is the fixed utility cost of moving, paid whenever $j' \neq j$; $I_{100}, I_{500}, \ldots$ are indicators for whether the distance between locations j and j' is higher than 100 km, 500 km, etc., so the remaining coefficients are increments for longer moves.[36]

$$\begin{aligned} \mu_t^e(j,j') \quad = \quad & \mu_{0,t}^e \, \mathbb{1}[j' \neq j] + \mu_{100,t}^e \, I_{100}(j,j') + \mu_{500,t}^e \, I_{500}(j,j') \\ & + \mu_{1000,t}^e \, I_{1000}(j,j') + \mu_{2000,t}^e \, I_{2000}(j,j') \end{aligned} \tag{28}$$

We estimate this first step by Maximum Likelihood with a Berry contraction mapping separately for each educational type e and year t, based on the choice probabilities in equation (14) (Bayer, Keohane and Timmins 2009, Mathes 2025). The maximization over the mean utilities is carried out inside the likelihood: for each candidate μ_t^e, the contraction $v_{j't}^e \leftarrow v_{j't}^e + \log s_{j't}^e - \log \hat{s}_{j't}^e(v_t^e, \mu_t^e)$, where $s_{j't}^e$ and $\hat{s}_{j't}^e$ are the observed and predicted shares of workers in location j', converges to the mean utilities that match the observed shares exactly. In the log-likelihood function, $I(j(i), j')$ is an indicator equal to 1 if the individual i who previously chose location j now chooses location j':

$$\max_{v_{j't}^e, \mu_t^e} \mathcal{LL}_t^e = \sum_{i \in N^e} I\left(j(i), j'\right) \, \log\left(p_t^e(j' \mid j(i))\right) \quad \forall e \in \{n, c\} \tag{29}$$

Analogous to the utility cost of moving, we estimate the mean utility of working in industry k, ξ_{kt}^e, as well as the utility cost of switching across industries δ_t^e. The switching costs

[36] For each metropolitan area in our sample, we find the geographic center from Census Bureau shapefiles and compute the pairwise distances in kilometers.

vary by educational group e and over time, and are parameterized as a function of the task distance between industries:

$$\begin{aligned}\delta^e_t(k,k') \;=\; & \delta^e_{0.05,t}\, I_{0.05}(k,k') + \delta^e_{0.3,t}\, I_{0.3}(k,k') + \delta^e_{0.8,t}\, I_{0.8}(k,k') \\ & + \delta^e_{1.5,t}\, I_{1.5}(k,k') + \delta^e_{2,t}\, I_2(k,k') + \delta^e_{4,t}\, I_4(k,k'),\end{aligned} \tag{30}$$

where $I_{0.05}, I_{0.3}, \ldots$ are indicator variables denoting whether the Euclidean task distance between industries k and k' is above 0.05, above 0.3, and so on.

The estimated cost functions are reduced-form objects: they measure the average utility cost of moving beyond a given distance, or of switching beyond a given task distance, net of wage and rent compensation, pooled across all sources of friction that scale with distance. For migration, at least three mechanisms plausibly contribute: literal relocation costs, including job and housing search at the destination; origin-specific amenities, such as family, social networks, and locally accumulated human capital, that are forgone upon moving; and the coordination costs of household-level moves when both partners hold place-specific labor market options. The data do not separate these channels, and we therefore report a single cost function per education group and year without attributing shares to specific mechanisms.

C Policy Counterfactuals

C.1 Welfare Measure

Welfare is computed from expected utility before the idiosyncratic taste shocks are realized. Let $\hat{v}^e_{j'k't}(j,k)$ denote the indirect utility in equation (3) net of the taste shock, evaluated at after-tax wages. Integrating over the Type I extreme value shocks, the expected utility of a worker of education e with origin cell (j,k) is

$$\bar{v}^e_{jkt} \;=\; \log \sum_{j',k'} \exp\big(\hat{v}^e_{j'k't}(j,k)\big).$$

With Cobb–Douglas preferences, raising consumption of goods and housing by x percent at given prices requires raising after-tax income by x percent, which raises utility by $\beta_w \log(1+x)$, since β_w is the utility weight on log income in equation (3). The consumption-equivalent variation of the policy for this worker is therefore

$$\mathrm{EV}^e_{jk} \;=\; \exp\left(\frac{\bar{v}^{e,1}_{jk} - \bar{v}^{e,0}_{jk}}{\beta_w}\right) - 1,$$

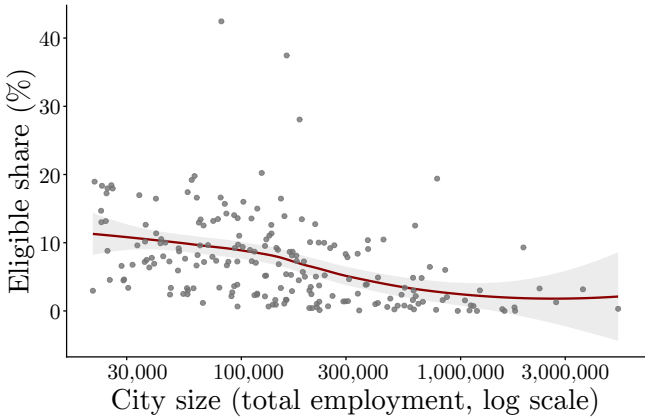
where superscripts 0 and 1 denote expected utility evaluated at the benchmark and at the counterfactual prices, wages, and taxes. The measure follows the convention of Diamond (2016). The reported welfare numbers aggregate EV^e_{jk} across origin cells using benchmark population weights, for the whole population or among eligible workers. Because expected

utility is computed at the origin, integrating over all destinations the worker may subsequently choose, the welfare gain of a worker whom the policy induces to move accrues to the pre-policy origin; eligibility is accordingly defined by the origin-cell wage in the benchmark. For the migration subsidy, the benchmark expected utility is evaluated without the subsidy in the choice set, so the gain includes the option value of the transfer as well as the cost of financing it.

C.2 Eligibility for the Migration Subsidy

Eligibility for the migration subsidy follows the income test of Section 6, applied at the level of the worker's own cell. Because low-wage cells are more common in smaller labor markets, the share of employment that qualifies falls with city size, from roughly 11 percent in the smallest markets to about 2 percent in the largest (Figure C1). Eligible workers are present throughout the city-size distribution; the subsidy reaches them only when they move into a targeted city.

Figure C1: Eligible share of employment by city size



Notes: Each point is one metropolitan area. The vertical axis is the share of the city's employment in cells whose benchmark wage is below 60 percent of the employment-weighted median wage. The horizontal axis is total city employment on a log scale. The curve is a local regression.

C.3 Shutting Off General-Equilibrium Channels

Table C1 solves each policy again with one general-equilibrium channel held at its benchmark value: agglomeration, the adjustment of the tax schedule that balances the government budget, pre-tax wages, rents, or the number of establishments. The housing policy acts on housing supply, so with rents held fixed it leaves the equilibrium unchanged and its welfare gain is zero whichever cities it targets.

The pre-tax wage response and entry move the gain from the housing policy in opposite directions depending on which cities it targets. Holding either fixed raises the gain when the policy targets New York, San Francisco, and San Jose, from 0.039 to 0.043 percent of consumption, and lowers it when it targets the concentrated cities, from 0.029 to 0.021 percent,

Table C1: Welfare with one general-equilibrium channel shut off

		Without				
	Full model	Agglomeration	Tax rebalancing	Wage response	Rent response	Firm entry
Housing policy						
New York, San Francisco, San Jose	0.03898	0.03931	0.03742	0.04338	0.00000	0.04268
Concentrated cities	0.02858	0.02772	0.02860	0.02126	0.00000	0.02118
Income-tested migration subsidy						
New York, San Francisco, San Jose						
Eligible workers	0.03739	0.03931	0.03807	0.03427	0.03098	0.04558
Whole population	−0.00282	−0.00256	−0.00215	−0.00017	−0.00262	−0.00047
Concentrated cities						
Eligible workers	0.11951	0.11902	0.12198	0.09945	0.11949	0.11953
Whole population	0.00637	0.00584	0.00879	0.00157	0.00664	0.00245

Notes: Welfare gains with labor market power in percent of consumption of goods and housing, averaged over the group shown with benchmark population weights; main calibration, 2019. Each column after the first holds one channel at its benchmark value; the wage column holds pre-tax wages. The first and last columns repeat the free-entry and entry-fixed columns of Table 4 for the smaller policy size. Eligible workers are defined as in Table 4.

the pattern that Section 6 traces to establishment entry at the targeted cities and exit elsewhere. Agglomeration and tax rebalancing change the gain by at most 4 percent.

For the migration subsidy, holding pre-tax wages fixed lowers the gain of eligible workers from 0.037 to 0.034 percent of consumption when the subsidy targets New York, San Francisco, and San Jose and from 0.120 to 0.099 percent when it targets the concentrated cities. These are reductions of 8 and 17 percent. Holding rents fixed lowers it by 17 percent at New York, San Francisco, and San Jose and by less than 1 percent at the concentrated cities. Agglomeration and tax rebalancing each change the gain of eligible workers by at most 5 percent. Holding the number of establishments fixed raises it by 22 percent when the subsidy targets New York, San Francisco, and San Jose, where exit no longer widens the markdowns of eligible workers, and leaves it unchanged when it targets the concentrated cities, where entry and exit offset each other (Appendix C.4). The whole-population effect is a small residual of the transfer and its financing. At New York, San Francisco, and San Jose it moves toward zero under every restriction and remains negative.

C.4 Where the Difference Between the Two Models Arises

Three results support the discussion of welfare in Section 6. First, establishment entry raises the welfare gain with labor market power for workers who live in the targeted cities and lowers it for workers who live elsewhere. Second, under the subsidy, the more elastic labor supply benefits mostly eligible workers outside the targeted cities, while establishment exit falls mostly on the markets where these workers work. Third, across four sets of targeted cities, the difference between the two models follows the markdown in the targeted cities relative to other cities.

Table C2 splits the difference between the two models in Table 4 by where workers live in the benchmark. Under the housing policy, moving from a fixed number of establishments

to free entry raises the difference for workers who live in the targeted cities, where establishments enter, and lowers it for those who live elsewhere, where they exit: by 2.4 and −11.1 percent of the gain without labor market power when the policy targets New York, San Francisco, and San Jose, and by 35.4 and −1.1 percent when it targets the concentrated cities. For eligible workers under the subsidy, the difference with the number of establishments held fixed accrues mostly to those who live outside the targeted cities: all of the 8.3 percent when the subsidy targets New York, San Francisco, and San Jose, and 9.3 of the 12.6 percent when it targets the concentrated cities.

Table C2: Difference between the models by where workers live

			Gain with labor market power relative to without (%)		
	Share of workers	Mean markdown	Entry fixed	Free entry	Free entry minus entry fixed
Housing policy	all				
New York, San Francisco, San Jose					
In the targeted cities	11.1	6.1	−0.2	2.2	2.4
Elsewhere	88.9	15.5	0.6	−10.5	−11.1
Total	100	14.5	0.4	−8.3	−8.7
Concentrated cities					
In the targeted cities	3.6	35.4	−3.1	32.3	35.4
Elsewhere	96.4	13.7	1.4	0.3	−1.1
Total	100	14.5	−1.7	32.6	34.3
Income-tested migration subsidy	eligible				
New York, San Francisco, San Jose					
In the targeted cities	0.7	2.3	0.0	0.0	0.0
Elsewhere	99.3	15.2	8.3	−11.2	−19.5
Total	100	15.1	8.3	−11.1	−19.5
Concentrated cities					
In the targeted cities	9.2	25.5	3.3	14.0	10.8
Elsewhere	90.8	14.1	9.3	−1.5	−10.8
Total	100	15.1	12.6	12.6	0.0

Notes: Main calibration, 2019; each policy at the 1× size of Table 4. Entry fixed and free entry are the welfare gain with labor market power less the gain without it, in percent of the gain without it of the workers of each block. Rows split this difference by where the workers live in the benchmark and add up to the total, which is the difference in Table 4. Share of workers and mean markdown, in percent, refer to the workers in each row, with benchmark employment weights. In the subsidy panel, rows refer to eligible workers, whose mean markdown is higher than that of all workers.

With the number of establishments held fixed, the labor supply elasticity of non-college cells below the income line outside the targeted cities rises by 0.04 percent when the subsidy targets New York, San Francisco, and San Jose and by 0.14 percent when it targets the concentrated cities, against close to zero in the other cells; under the housing policy the two groups of cells move alike. Under free entry, the workers the subsidy draws away leave the markets where eligible workers work. When the subsidy targets New York, San Francisco, and San Jose, the number of establishments in non-college cells below the income line outside the three cities falls by 0.07 percent relative to the equilibrium with the number of establishments

held fixed, against 0.02 percent in the other cells outside the three cities, and their markdown widens by 0.007 percentage points, against 0.002. Since 0.7 percent of eligible workers live in the three cities, the change from entry for eligible workers is −19.5 percent. When the subsidy targets the concentrated cities, the same comparison gives 0.16 against 0.01 percent. There, the 9.2 percent of eligible workers who live in the targeted cities gain from entry, by 10.8 percent of the gain without labor market power, about as much as those elsewhere lose from exit.

Table C3 and Figure C2 repeat both policies for two further sets of targeted cities. Ranking cities by their employment-weighted Herfindahl index, each set is the group of cities that starts at a given percentile of cumulative employment, the 50th and the 20th, and holds the same share of employment as the concentrated cities. In both sets, the number of establishments held fixed leaves the housing policy's gain within 1 percent of the gain without labor market power, and raises the subsidy's gain for eligible workers by 9.7 and 15.3 percent. Under free entry, the difference has the sign of the gap between the markdown in the targeted cities and that in the rest of the country for both policies, and rises with this gap. Interpolating between the four sets, the difference between the two models is zero at a markdown gap of about 0.1 percentage points for the housing policy and about 2 percentage points for the subsidy, where the more elastic labor supply and establishment exit offset each other (Figure C2).

Table C3: Difference between the models across four sets of targeted cities

	Mean markdown (%)		Housing policy		Migration subsidy	
	Targeted	Other	Entry fixed	Free entry	Entry fixed	Free entry
New York, San Francisco, San Jose	6.1	15.5	0.4	−8.3	8.3	−11.1
Four large cities	11.4	14.6	0.5	−3.8	9.7	−5.8
Eleven medium-sized cities	21.0	14.3	−0.6	7.3	15.3	5.4
Concentrated cities	35.4	13.7	−1.7	32.6	12.6	12.6

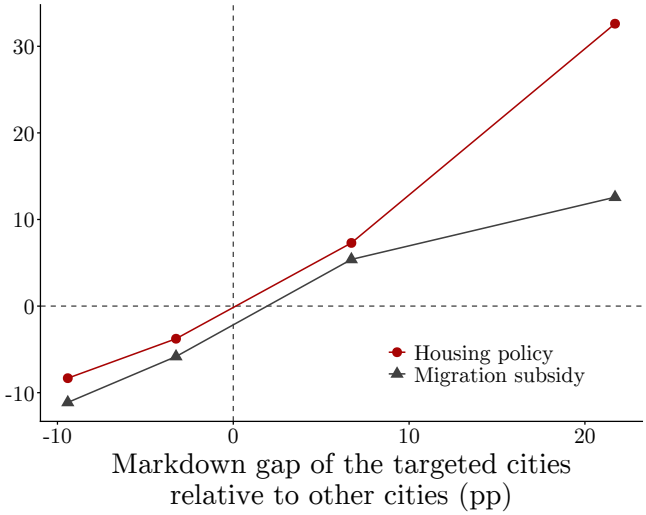
Notes: Main calibration, 2019, each policy at its 1× size. Mean markdowns weight location-industry cells by benchmark employment, over the targeted cities and over all other cities. The policy columns report the welfare gain with labor market power less the gain without it, in percent of the gain without it, for all workers under the housing policy and for eligible workers under the subsidy. Ranking cities by the employment-weighted Herfindahl index, the two middle sets start at the 50th and the 20th percentile of cumulative employment and hold the same share of employment as the concentrated cities. The four large cities are Baltimore, Portland, West Palm Beach, and Providence; the eleven medium-sized cities are Richmond, Louisville, Rochester, Omaha, Albuquerque, Little Rock, McAllen, Wilmington, Bakersfield, Springfield, and Santa Rosa.

C.5 Wage Decomposition: Marginal Product versus Markdown

The pre-tax wage in the model is $w^e_{jk} = \text{MPL}^e_{jk} \cdot (1 - \psi^e_{jk})$, where ψ^e_{jk} is the markdown, so any counterfactual wage change decomposes exactly in logs:

$$\Delta \log w^e_{jk} = \underbrace{\Delta \log \text{MPL}^e_{jk}}_{\text{marginal product}} + \underbrace{\Delta \log(1 - \psi^e_{jk})}_{\text{markdown}}. \tag{31}$$

Figure C2: Welfare gain with labor market power relative to without (%)



Notes: Each point is one set of targeted cities (Table C3). The vertical axis is the welfare gain with labor market power and free entry less the gain without labor market power, in percent of the latter, for all workers under the housing policy and for eligible workers under the subsidy. The horizontal axis is the markdown gap: the employment-weighted mean markdown in the targeted cities minus that in all other cities, in percentage points. Main calibration, 2019.

The first term is the wage change that would obtain if firms were price-takers: diminishing returns, education-mix changes through the CES aggregator, and agglomeration. The second term is the part of the wage change that comes from the markdown. Unlike the exercises of Appendix C.3, which solve the equilibrium again with a channel held fixed, equation (31) is an accounting identity at the realized equilibrium.

Table C4 reports the decomposition for the housing policy. At the concentrated cities, the markdown term accounts for most of the wage gain: 0.28 of the 0.31 percent rise in the pre-tax wage of non-college workers, and 0.21 of 0.26 percent for college workers. At New York, San Francisco, and San Jose both terms are small; in San Francisco, where employment rises the most, the markdown term contributes 0.032 of the 0.056 percent non-college wage gain. In the cities that lose the most workers the markdown widens, and its term is negative in every city listed.

C.6 Untargeted Moments

We compute both responses compared with published estimates in Section 4.3 from the housing policy of Section 6 at New York, San Francisco, and San Jose, under the main calibration, in the 2019 stationary equilibrium.

Population and the Real Wage Since $\beta_r = -\alpha\beta_w$, wages and rents enter indirect utility (3) only through the housing-adjusted real wage, $\log \tilde{w}_j = \log y_j - \alpha \log r_j$. A city's population therefore responds to prices as

$$\Delta \log N = \varepsilon_M \left(\Delta \log y - \alpha \, \Delta \log r \right), \tag{32}$$

Table C4: Wage decomposition for the housing policy

	Non-college				College			
	$\%\Delta N$	$\Delta \log w$	$\Delta \log \text{MPL}$	$\Delta \log(1-\psi)$	$\%\Delta N$	$\Delta \log w$	$\Delta \log \text{MPL}$	$\Delta \log(1-\psi)$
Targeting New York, San Francisco, San Jose								
New York, NY-Northeastern NJ	0.392	0.017	0.010	0.007	0.370	0.030	0.018	0.011
San Francisco-Oakland-Vallejo, CA	0.758	0.056	0.023	0.032	0.738	0.070	0.033	0.038
San Jose, CA	0.117	0.008	−0.001	0.009	0.098	0.019	0.006	0.013
Rest of the country, mean	−0.044	−0.012	−0.006	−0.005	−0.070	−0.005	0.002	−0.007
Madison, WI	−0.053	−0.019	−0.008	−0.011	−0.086	−0.018	0.000	−0.018
Ann Arbor, MI	−0.051	−0.018	−0.008	−0.009	−0.084	−0.016	0.000	−0.016
Wichita, KS	−0.060	−0.019	−0.006	−0.013	−0.086	−0.014	0.001	−0.015
Lafayette-W. Lafayette, IN	−0.059	−0.022	−0.005	−0.017	−0.085	−0.017	0.002	−0.019
Chattanooga, TN/GA	−0.055	−0.020	−0.006	−0.014	−0.083	−0.016	0.001	−0.018
Targeting the concentrated cities								
Concentrated cities (31), mean	1.056	0.311	0.034	0.277	0.890	0.256	0.047	0.209
Rest of the country, mean	−0.046	−0.001	0.002	−0.003	−0.024	−0.007	−0.005	−0.002
Wichita, KS	−0.058	−0.007	0.002	−0.009	−0.033	−0.012	−0.006	−0.006
Stockton, CA	−0.054	−0.006	0.000	−0.007	−0.031	−0.013	−0.007	−0.006
McAllen-Edinburg-Pharr-Mission, TX	−0.055	−0.005	0.001	−0.006	−0.031	−0.011	−0.006	−0.004
Tulsa, OK	−0.055	−0.005	0.001	−0.007	−0.031	−0.011	−0.006	−0.005
Bakersfield, CA	−0.052	−0.006	0.001	−0.006	−0.029	−0.012	−0.007	−0.005

Notes: Percent changes; main calibration, 2019, free entry. The last three columns of each block are the terms of equation (31). City entries are benchmark-employment-weighted means of cell-level log changes. The last five rows of each panel are the cities outside the targeted set with the largest percentage fall in employment among cities larger than the median.

where ε_M is the elasticity we compare. We solve the policy with pre-tax wages, the tax schedule, agglomeration, and the number of establishments held at their benchmark values, so that $\Delta \log y = 0$ and rent is the only price that moves. The policy shifts housing supply and leaves equation (32) unchanged, so the ratio of the two equilibrium changes identifies $\varepsilon_M = -\Delta \log N / (\alpha \, \Delta \log r)$ independently of the slope of housing supply and of the size of the policy. Averaging the population and rent changes over the three cities with benchmark population weights, the model gives $\varepsilon_M = 1.23$ (Table C5).

Reduced-form studies report the responses of population, earnings, and rents to a single local shifter, b_N, b_y, and b_r. Substituting them into equation (32) gives $\varepsilon_M = b_N / (b_y - \alpha \, b_r)$, which we evaluate at $\alpha = 0.30$. Unlike in our experiment, earnings move in these designs, so the earnings term cannot be dropped: a labor demand shifter raises earnings, the workers it draws in raise rents, and the ratio b_N / b_r alone would measure the slope of housing supply rather than ε_M. Published values vary widely across designs. Table C5 reports three estimates for which every underlying coefficient is individually significant: the conversions of Allcott and Keniston (2018) and Hornbeck and Moretti (2024), and the structural estimate of Suárez Serrato and Zidar (2016). The model's 1.23 lies at the lower end of these, next to the 1.21 of Allcott and Keniston (2018) and the 1.12 of Suárez Serrato and Zidar (2016).

Establishments and Local Employment The number of establishments in the model is the number of symmetric-equivalent employers, $E_{jk} = 1/\text{HHI}_{jk}$. We solve the policy with all channels active and free entry, and regress $\Delta \log E$ on $\Delta \log N$ across location-industry cells through the origin, weighting by benchmark employment. The elasticity is 0.58. Published estimates count establishments instead. For M establishments whose employment has coef-

ficient of variation CV, HHI $= (1 + CV^2)/M$, so

$$\frac{d \log M}{d \log N} = \frac{d \log E}{d \log N} + \frac{d \log(1 + CV^2)}{d \log N}. \tag{33}$$

The model has a single establishment size in each cell, so we estimate the second term on County Business Patterns, across metropolitan area by industry cells with industry fixed effects. It is 0.20 on average and rises with the national concentration of the industry, from -0.02 in the least concentrated quartile of industries to 0.37 in the most concentrated.[37] Adding each cell's quartile term before aggregating gives an elasticity of the establishment count of 0.79 (Table C5). In the 2016 cross-section of the same data, the number of establishments rises by 0.91 percent for each percent of employment across metropolitan area by industry cells with industry fixed effects, and by 0.98 percent across metropolitan area totals. Published designs divide the establishment response to a labor supply shock by the employment or population response to the same shock. The ratio is 0.26 for immigration into U.S. commuting zones over nine years (Mahajan 2024). For the settlement of expellees in West German counties after 1945, it is 0.57 by ordinary least squares and 1.15 with instrumental variables in the short run, up to 1950, and 0.96 and 1.71 in the long run, up to 1956–61 (Peters 2022).

Table C5: Untargeted elasticities in the model and in the data

	Model	Data
Population to the housing-adjusted real wage		
This model, housing policy	1.23	
Suárez Serrato and Zidar (2016), state corporate taxes		1.12
Allcott and Keniston (2018), oil and gas booms		1.21
Hornbeck and Moretti (2024), manufacturing productivity		4.12
Number of establishments to local employment		
This model, housing policy	0.79	
County Business Patterns, metropolitan areas, 2016		
Within industry		0.91
Metropolitan area totals		0.98
Mahajan (2024), immigration		0.26
Peters (2022), post-war refugee inflows		
Short run, OLS		0.57
Short run, IV		1.15
Long run, OLS		0.96
Long run, IV		1.71

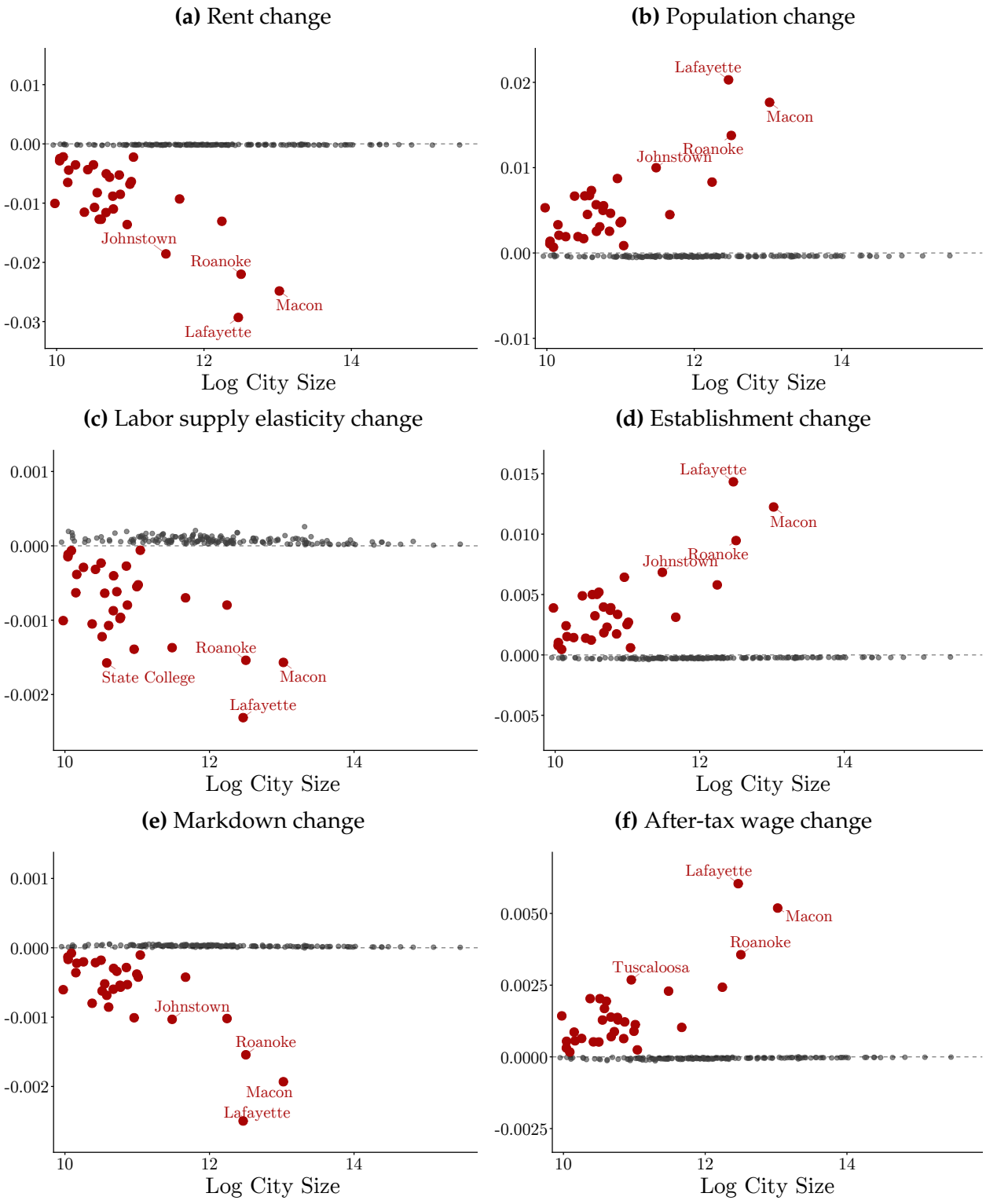
Notes: Entries are computed and converted as described in the text. County Business Patterns: employment-weighted regressions across metropolitan area by industry cells with industry fixed effects (within industry) or across metropolitan area totals. Peters (2022): West German counties, short run to 1950, long run to 1956–61.

[37] The quartiles are of national concentration rather than of the cell's own Herfindahl index, because the index enters both E and the dispersion term.

C.7 Additional Policy Cross-Sections

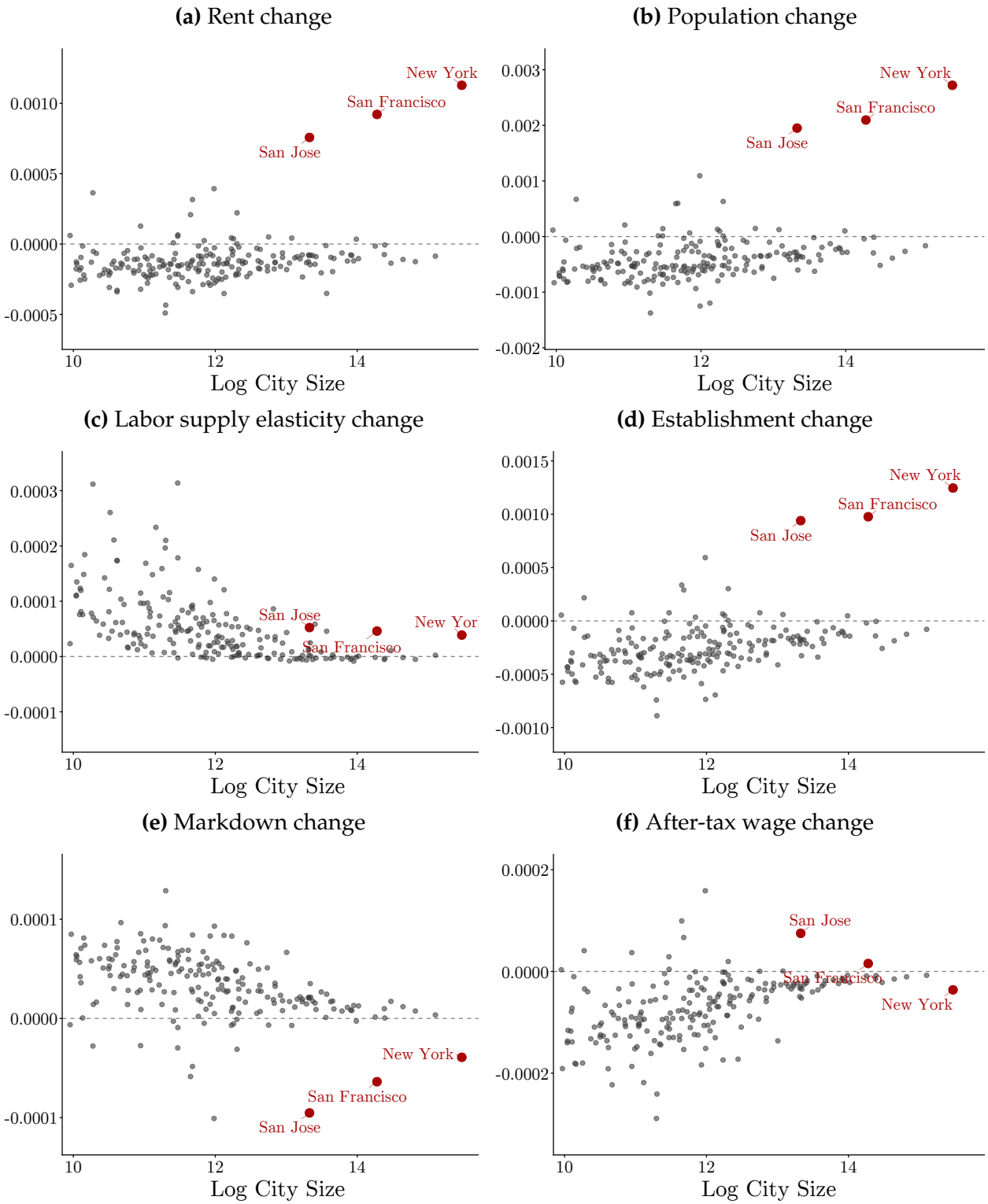
Figures C3 and C4 report the city-level cross-sections referenced in Section 6: the housing policy at the concentrated cities and the migration subsidy directed at New York, San Francisco, and San Jose.

Figure C3: Housing policy at the concentrated cities



Notes: Conventions as in Figure 3; concentrated cities in red.

Figure C4: Migration subsidy directed at New York, San Francisco, and San Jose



Notes: Conventions as in Figure 3; New York, San Francisco, and San Jose in red.

D Shift-Share Estimate and Shift-Share Calibration

This appendix reports a shift-share estimate of the wage coefficient β_w and repeats the counterfactuals of Section 6 with β_w set to it. The counterfactual predictions depend on β_w and on the moving and switching costs. An error in these parameters would leave the fit to the

data intact, because the amenities ξ^e_{jkt} absorb it, but would distort the reallocation the model predicts in response to the policies we study.

D.1 Specification

Given the mean utilities $v^e_{j't}$ estimated in the first step (Section 4.2), the wage and rent coefficients satisfy, in differences that eliminate time-invariant unobservables,

$$\Delta v^e_{j't} = \beta_w \Delta \log(y^e_{j't}) + \beta_r \Delta \log(r_{j't}) + \Delta \xi^e_{j't} \quad \forall e \in \{n, c\} \tag{34}$$

We estimate the wage coefficient β_w in equation (34), which determines the labor supply elasticity and hence the degree of labor market power (equations 6 and 9); the rent coefficient is tied to β_w through a calibrated housing expenditure share, as explained below.[38] A typical concern when estimating equation (34) is that time-varying unobserved location-specific amenities may be correlated with changes in rents and wages, leading to omitted variable bias in the estimation of β_w and β_r. To address this issue and isolate plausibly exogenous variation in wages and rents over time, we construct a novel shift-share instrument based on changes in labor demand induced by import competition following the Canada–U.S. and North American Free Trade Agreements (CUSFTA and NAFTA).

CUSFTA and NAFTA, implemented in 1989 and 1994, respectively, reduced trade barriers between Canada, the United States, and later Mexico. These agreements triggered a sharp expansion in cross-border trade flows beginning in the late 1980s, and altered the composition of labor demand across industries with differential exposure to import competition. Figure D1 illustrates this evolution by plotting how the log value of imports from Canada and Mexico (relative to imports from the rest of the world) evolves over time. Before 1988, there is no systematic difference, but following the implementation of CUSFTA and especially NAFTA there is a clear and persistent increase in the relative value of imports from these partner countries. The timing of these institutional shocks therefore provides plausibly exogenous time-series variation that we exploit to construct our shift-share instrument.

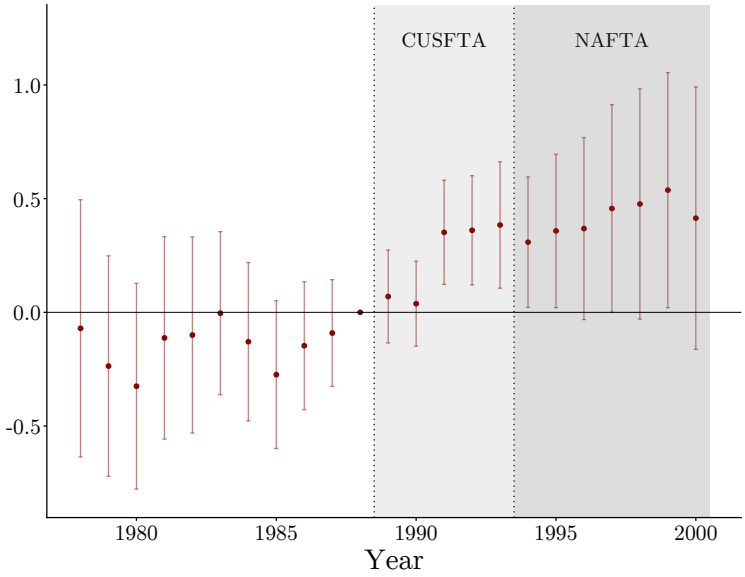
The instrument combines 1980 industry employment shares in location j and education group e with changes in the industry-level log value of imports from Canada and Mexico between 1980 and 2000:

$$Z_{je} = \sum_{k=1}^{K} s_{ejk,1980} \, \Delta \text{log Imports}_{k,1980-2000}$$

We use predetermined 1980 base shares, which mitigates concerns that local employment composition may itself respond to contemporaneous trade shocks. The "shift" component

[38] Because establishments interact strategically within each local labor market, the wage and employment responses of a single firm to its own shocks understate the labor supply elasticity it faces, as its competitors respond (Berger, Herkenhoff and Mongey 2022, p. 1148). A shock common to all firms in a market does not have this problem and identifies the market-level labor supply elasticity (Berger, Herkenhoff and Mongey 2022, p. 1167). Equation (34) uses variation across locations over time induced by shocks of this kind.

Figure D1: Log value of imports from Canada and Mexico relative to the rest of the world



Notes: The figure plots year-by-treatment fixed effects from a regression where the outcome variable is the log value of imports. The treatment is defined as imports originating from Canada and Mexico; the control group is imports from the rest of the world. The specification includes year, treatment, and country fixed effects, 1978–2000, with 1988 as the reference year. Error bars are 90% confidence intervals from heteroskedasticity-robust standard errors. Data: Schott (2008), U.S. Census Bureau.

comes from long-run changes in industry-level import values from Canada and Mexico induced by the trade liberalizations, while the "share" component captures differential initial exposure across local labor markets. The exposure measure is related to the local exposure measures of Hakobyan and McLaren (2016) and Choi et al. (2024), who use them directly as regressors for local wages, employment, and voting. We measure the shifts by changes in imports from both Canada and Mexico, and use the exposure to instrument the wage and rent changes that enter workers' location choices.

Equation (34) contains two endogenous prices, wages and rents, while the instrument provides a single source of exogenous variation. We therefore impose the housing expenditure share $\alpha = 0.30$ of Section 4.3 and estimate a single coefficient on the housing-adjusted real wage $\tilde{w}^e_{jt} \equiv y^e_{jt} r^{-\alpha}_{jt}$, where y^e_{jt} is the after-tax location wage, with the rent coefficient recovered as $\beta_r = -\alpha\beta_w$. Consistent with the time frame of the instrument, we estimate the long-difference specification between 1980 and 2000 with a single endogenous regressor:

$$v^e_{j2000} - v^e_{j1980} = \beta_w(\log \tilde{w}^e_{j2000} - \log \tilde{w}^e_{j1980}) + \gamma' \mathbf{x}_{je} + \epsilon_{je} \tag{35}$$

where $\mathbf{x}_{je}$ collects college fixed effects, the amenity index based on PCA (Diamond 2016), and the local sum of exposure shares discussed next.

The exogeneity of the instrument rests on the shifts rather than the shares (Borusyak, Hull and Jaravel 2024). Because the shares may be correlated with local unobservables, for example through past labor reallocation across industries, we control for the local sum of exposure shares, as Borusyak, Hull and Jaravel (2024) recommend, so that identification comes only from the quasi-random national component of the trade shocks. Inference accounts for the

shift-share structure as well: the standard error on $\hat{\beta}_w$ is exposure-robust at the industry level, computed from the equivalent shock-level regression of Borusyak, Hull and Jaravel (2024).

To support the plausibility of the exogeneity assumption for the industry-level shifts, we first note the absence of differential pre-trends in log import values from Canada and Mexico compared to the rest of the world prior to the trade liberalizations (Figure D1). Moreover, Appendix Table D3 documents that the national import shocks are not systematically related to changes in industry-level worker composition by age, gender, or race.

Local Amenities Observable amenities are included as controls in the specification above. We measure amenities at the location–year level following Diamond (2016). Information on amenities related to retail, transportation, innovation, crime, environmental quality, and schooling quality is drawn from multiple sources, including the U.S. Census, the County Business Patterns, the FBI Uniform Crime Reports, the EPA Air Quality Monitoring System, the Census of Governments, the NBER Patent Database, the U.S. Patent and Trademark Office, and Duranton and Turner (2011). Retail amenities capture the variety of shops and entertainment options in each city, measured by the per capita number of clothing stores, eating and drinking places, and cinemas. Transportation amenities reflect the quality of public transit and road infrastructure, including indicators such as buses per person, an overall public transit rating, and average daily traffic on major roads. Innovation activity is proxied by the number of patents per capita. Crime data include both violent and property crimes per person. Environmental quality is measured by per capita government spending on parks and recreation and by the EPA air quality index. School quality indicators include per-student government spending on K–12 education and the ratio of students enrolled in private to public schools.

A composite amenity index is then constructed for each location and year using principal component analysis (PCA), following Diamond (2016). Unlike Diamond, we extend the index to 2019. Because transportation data are unavailable for that year, we use 2010 values as proxies for transportation amenities.

Trade Data Industry-level import data used for the shift-share IV are drawn from Schott (2008), who compiled detailed trade information from the U.S. Census Bureau.

D.2 Estimates

Table D1 reports the estimates. The OLS coefficient in column (1) is positive but small. This is the attenuation expected when unobserved amenity improvements draw workers into a location and depress its wage conditional on labor demand: the migration response to observed wage changes then understates the underlying utility coefficient. The IV estimate in column (2) is $\hat{\beta}_w = 5.09$, with an exposure-robust standard error of 3.01 and a first-stage F of 18.3. The magnitude is consistent with the shift-share estimates of Diamond (2016), which range from 3.26 to 4.98 in comparable specifications, although the identifying variation differs (import-driven shifts rather than national industry growth rates). The implied rent coefficient

Table D1: OLS and IV estimates of labor supply parameters

		Δ Mean Utility Locations	
		(1)	(2)
$\Delta \log \tilde{w}$		0.878***	5.087*
		(0.215)	(3.006)
Controls:	College FEs	✓	✓
	City Amenities	✓	✓
	Local Sum of Shares	✓	✓
Observations		438	438
Estimation Method		OLS	IV
F-test (First Stage, $\Delta \log \tilde{w}$)		–	18.29

Notes: The table reports the OLS and IV estimates of β_w from the second-step location-choice regression $\Delta v_{j,c} = \beta_w \Delta \log \tilde{w}_{j,c} + \gamma' \mathbf{x}_{j,c} + \varepsilon_{j,c}$, where $\tilde{w} \equiv w \cdot r^{-\alpha}$ is the housing-adjusted real wage with $\alpha = 0.30$. The IV is a 1980–2000 NAFTA-shock shift-share on 1980 MSA-industry employment shares, with the local sum of exposure shares as a control (Borusyak, Hull and Jaravel 2024). All specifications include college fixed effects, city amenities, and the local sum of shares. The OLS standard error in column (1) is heteroskedasticity-robust at the observation level; the IV standard error in column (2) is exposure-robust at the shock (IND1990) level, computed from the equivalent shock-level regression of Borusyak, Hull and Jaravel (2024, Proposition 5). *p<0.1, **p<0.05, ***p<0.01.

is $\beta_r = -\alpha\hat{\beta}_w = -1.53$.

D.3 Diagnostics

Several diagnostics support the design; the underlying tables and figures are collected at the end of this subsection. First, a placebo regression replaces the outcome with the post-period change in mean utility between 2000 and 2010, keeping the endogenous regressor, the instrument, and the controls at the 1980–2000 window, so that the first stage is identical to the main specification. Under the exclusion restriction the placebo coefficient should be indistinguishable from zero; the estimate is statistically insignificant (Appendix Table D2).

Second, the identifying variation is broad based. The effective shock-level sample size, the inverse Herfindahl index of aggregate industry exposure shares (Borusyak, Hull and Jaravel 2024), is approximately 44, and the Rotemberg weights (Goldsmith-Pinkham, Sorkin and Swift 2020) are dispersed: the largest single-industry weight is 0.24 and the top ten industries account for about two thirds of the absolute weight (Appendix Table D4). Re-estimating the specification dropping one industry at a time barely moves the estimate (Appendix Figure D2); in particular, the top-weighted industry, guided missiles, has no usable first stage of its own, and dropping it changes the estimate very little. Appendix Figure D3 plots the residualized first stage and reduced form underlying the estimate.

Third, the balance diagnostics clarify which identifying assumption the design can rest on. Under the share-based reading of Goldsmith-Pinkham, Sorkin and Swift (2020), each 1980 exposure share serves as its own instrument, which requires exposure to be balanced with respect to predetermined location characteristics. This condition fails in our setting: more exposed locations had systematically lower mean utility, wages, and rents in 1980, reflecting their manufacturing intensity (Appendix Tables D5 to D7). Identification therefore

rests on the exogeneity of the national import shifts, with the sum-of-shares control so that identification comes only from their quasi-random component. Level differences across locations drop out of the long-difference specification by construction; the remaining concern is a differential trend correlated with exposure, which the placebo regression, the absence of import pre-trends, and the balance of the shifts on industry composition address.

Table D2: Placebo test of the shift-share instrument

		Δ Mean Utility Locations
		(1)
$\Delta \log \tilde{w}$		0.599
		(1.844)
Controls:	College FEs	✓
	City Amenities	✓
	Local Sum of Shares	✓
Observations		438
Estimation Method		IV
F-test (First Stage, $\Delta \log \tilde{w}$)		18.29

Notes: Placebo test: the outcome is the change in mean location utility between 2000 and 2010, while the endogenous regressor $\Delta \log \tilde{w}$ (the housing-adjusted real wage) and the shift-share IV are both built on the 1980–2000 window. The first stage is therefore identical to Table D1. Standard errors are exposure-robust at the shock (IND1990) level (Borusyak, Hull and Jaravel 2024, Proposition 5). *p<0.1, **p<0.05, ***p<0.01.

Table D3: Relationship between the NAFTA shift and industry-level observables

	Δ Mean Age	Δ Male Share	Δ Black Share
	(1)	(2)	(3)
Shift	-0.0561	-0.0010	-0.0001
	(0.0792)	(0.0023)	(0.0012)
College FEs	✓	✓	✓
Observations	128	128	128

Notes: Each column is a separate regression of the indicated change in industry-level observable (1980–2000) on the shift-share instrument, with college fixed effects. The outcomes are limited to predetermined demographic characteristics. Heteroskedasticity-robust standard errors in parentheses. *p<0.1, **p<0.05, ***p<0.01.

D.4 Robustness

Two robustness checks vary the sample and the measurement window. The first addresses the concern that product market power in locally sold output could bias the wage coefficient: we restrict the sample to traded industries (Delgado, Bryden and Zyontz 2014), whose output prices are largely set outside the local market, and obtain $\hat{\beta}_w = 4.65$ (s.e. 4.25) on a subsample covering 113 of the 177 industries and about half of employment. Because the location wage has to be recomputed from the microdata for this subsample, the comparable all-industry estimate recomputes it the same way and is 5.01 (2.96), close to the 5.09 of Table D1. The traded-industry estimate is close to both but imprecise. The second measures migration flows over a one-year window rather than the five-year window of Table D1, with the migration

Table D4: NAFTA shift-share: ten largest industries by Rotemberg weight

Industry	Weight $\hat{\alpha}_k$	Shift g_k	IV estimate $\hat{\beta}_k$	First-stage F_k
Guided missiles	0.241	−7.17	−18.96	0.3
Other electrical machinery	0.124	3.03	−0.90	14.4
Apparel	0.122	3.64	3.80	7.8
Computers and peripherals	0.115	3.25	−0.59	13.7
Plastics and synthetics	−0.093	3.28	−4.81	7.6
Medical and dental instruments	0.091	3.76	0.55	20.4
Other yarn and fabric	0.081	3.62	4.24	1.3
Radio, TV and communication	0.065	3.79	0.43	1.0
Blast furnaces and steel	0.043	1.31	−0.31	9.5
Knitting mills	0.038	3.53	7.67	5.2

Notes: Rotemberg weights $\hat{\alpha}_k$ (Goldsmith-Pinkham, Sorkin and Swift 2020, Proposition 3.1) for the place-level composite IV specification, computed over the 64 IND1990 industries with a 1980–2000 import shift; weights sum to one ($\sum_{\alpha_k>0}\hat{\alpha}_k = 1.27$, $\sum_{\alpha_k<0}\hat{\alpha}_k = -0.27$). g_k is the industry shift, $\hat{\beta}_k$ the just-identified IV estimate using only industry k's shift-share instrument, and F_k its first-stage F.

Table D5: NAFTA shift-share: balance on 1980 levels

1980 outcome	Coefficient on Z_j	(SE)
Mean utility $v^e_{j,1980}$	−0.691***	(0.249)
Log net wage	−0.131***	(0.031)
Log rent	−0.200***	(0.050)
Amenity index	−1.534*	(0.822)

Notes: Each row reports a separate regression of a 1980 (pre-shift) outcome level on the aggregate NAFTA shift-share instrument Z_j, with the local sum of exposure shares as a control (Borusyak, Hull and Jaravel 2024) and an education fixed effect. Heteroskedasticity-robust standard errors in parentheses. *p<0.1, **p<0.05, ***p<0.01.

Table D6: NAFTA shift-share: share-level balance diagnostics

Statistic	Value
Rotemberg-weighted $\sum_k \hat{\alpha}_k\hat{\beta}_k$	4.559
(SE; p-value)	(1.141; 0.00)
Individual rejection rate at $p < 0.05$	13/64 (20%)
Individual rejection rate at $p < 0.10$	19/64 (30%)
Correlation of $\hat{\alpha}_k$ and $\hat{\beta}_k$	0.087

Notes: Share-level balance regressions follow Goldsmith-Pinkham, Sorkin and Swift (2020, Section 5). The 1980 level of location mean utility $v^e_{j,1980}$ is regressed separately on each of the $K = 64$ share-level instruments $s_{jk,1980}$, with an education fixed effect. Because 1980 is the first sample year, balance is evaluated on predetermined levels rather than pre-period changes. Rotemberg weights $\hat{\alpha}_k$ are those of Table D4; the Rotemberg-weighted standard error is computed under an independence approximation across industries.

costs and mean utilities re-estimated on that window, and yields $\hat{\beta}_w = 4.44$ (s.e. 2.20, first-stage $F = 18.3$), close to the five-year estimate. All standard errors are exposure-robust.

D.5 Markdown Decline under the Shift-Share Calibration

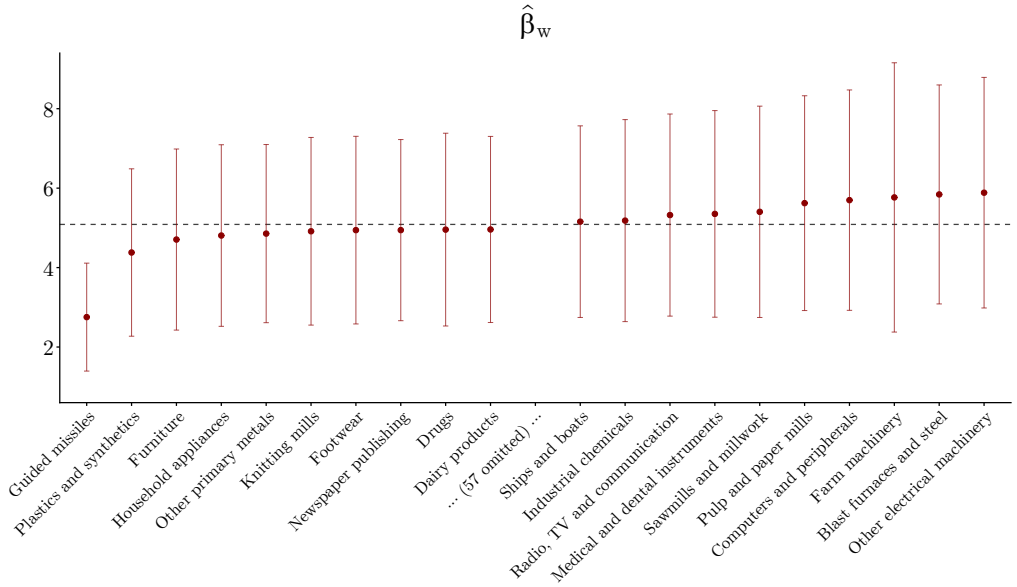
Markdowns are roughly a quarter as large under the shift-share calibration as under the main calibration. The employment-weighted mean markdown falls from 8.2 percent in 1980 to 3.0 percent in 2019, a decline of 63 percent (Figure D4). The Shapley attribution ranks the

Table D7: Share-level balance for the ten largest industries by Rotemberg weight

Industry	Weight $\hat{\alpha}_k$	Balance coefficient $\hat{\beta}_k$	(SE)
Guided missiles	0.241	15.37***	(2.86)
Other electrical machinery	0.124	−0.50	(1.56)
Apparel	0.122	−4.60*	(2.56)
Computers and peripherals	0.115	9.12***	(2.36)
Plastics and synthetics	−0.093	−2.03	(2.29)
Medical and dental instruments	0.091	7.96	(5.98)
Other yarn and fabric	0.081	−2.21	(1.52)
Radio, TV and communication	0.065	4.46**	(1.98)
Blast furnaces and steel	0.043	−1.13	(1.31)
Knitting mills	0.038	−6.83	(5.95)

Notes: Individual share-level balance coefficients $\hat{\beta}_k$ from separate regressions of the 1980 balance outcome ($v^e_{j,1980}$, education fixed effect included) on the exposure share $s_{jk,1980}$, for the ten industries with the largest Rotemberg weights. *p<0.1, **p<0.05, ***p<0.01.

Figure D2: Leave-one-out distribution of $\hat{\beta}_w$



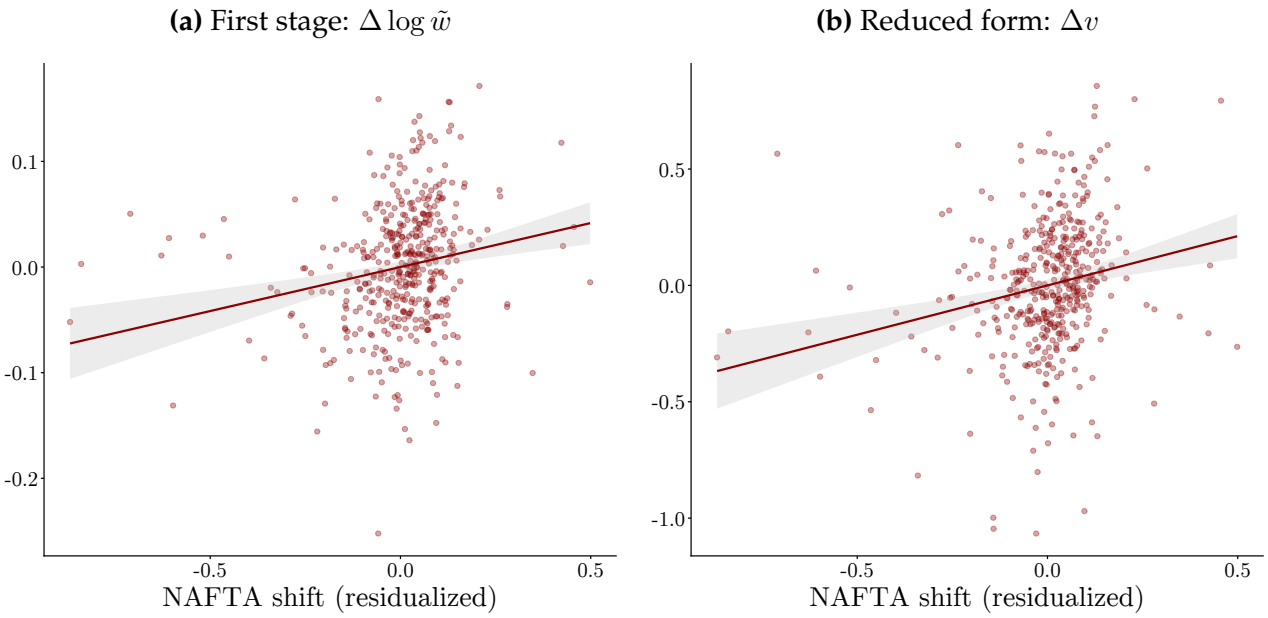
Notes: Each point is the IV estimate of equation (35) with one 3-digit Census industry removed from the shift; the figure shows the ten industries whose removal moves $\hat{\beta}_w$ the most in each direction, with the remaining industries collapsed to the central marker. Error bars are 90% confidence intervals from observation-level heteroskedasticity-robust standard errors; the dashed line marks the estimate from the full shift.

channels as under the main calibration: establishment counts account for 34 percent of the decline, falling industry-switching costs for 31 percent, and rising migration costs offset 9 percent, so that the three channels explain 57 percent of it (Table 3).

D.6 Counterfactuals under the Shift-Share Calibration

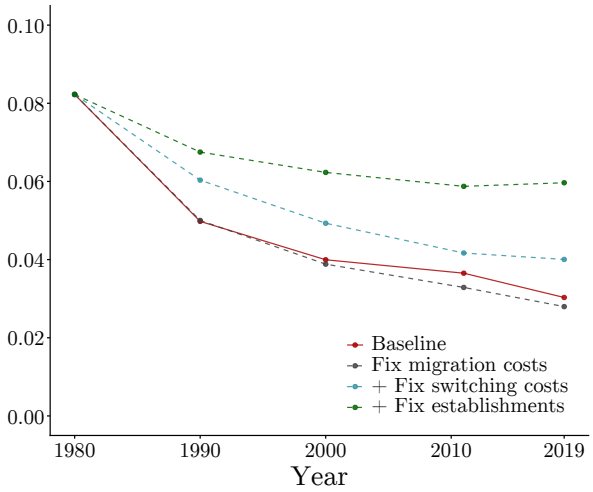
With β_w set to the shift-share estimate, markdowns are smaller than under the main calibration, about 4 percent against 18 percent in the employment-weighted mean pooled over cells and years. This markdown corresponds to a firm-level labor supply elasticity of about 22. Published estimates of this elasticity depend on the identifying variation: in the meta-

Figure D3: Residualized first stage and reduced form



Notes: Scatter plots of the residualized dependent variable against the residualized shift-share instrument after partialling out college fixed effects, the amenity index, and the local sum of exposure shares. The solid line is the OLS fit with a 95% confidence band.

Figure D4: Mean markdowns across location-industries, shift-share calibration



Notes: Conventions as in Figure 2, with the wage coefficient at its shift-share estimate.

analysis of Sokolova and Sorensen (2021), studies that identify it from the inverse labor supply elasticity average 22.8, against 4.5 across studies published in leading journals, the target of the main calibration.

Under the shift-share calibration we hold the number of establishments at its benchmark value in every policy equilibrium, because with free entry some markets have no interior equilibrium.[39] The column with labor market power in Table D8 is therefore comparable to

[39] A wage increase draws workers into a market, the inflow attracts establishments, and entry narrows the markdown and raises the wage again. Free entry has an interior equilibrium only if the gain of this loop, $\beta_w \cdot \psi \cdot \varepsilon_m$, is below one in every market, where ψ is the markdown and ε_m the elasticity of the number of establishments to employment. Its largest value is 0.37 under the main calibration. Under the shift-share calibration it exceeds one in markets that account for 0.15 percent of national employment and about 2 percent of

Table D8: Welfare effects of the housing policy and the migration subsidy, shift-share calibration

		Labor market power		Difference	
		Without	With	Points	Percent
Housing policy, 1×: inverse supply elasticity 4.7 percent lower					
New York, San Francisco, San Jose					
1×		0.0399	0.0401	0.0002	0.4
2×		0.0848	0.0852	0.0004	0.4
Concentrated cities					
1×		0.0249	0.0242	−0.0007	−2.8
2×		0.0521	0.0506	−0.0015	−2.9
Income-tested migration subsidy, 1×: \$5,500 per mover					
New York, San Francisco, San Jose					
1×	Eligible workers	0.0766	0.0819	0.0053	6.9
	Whole population	−0.0039	−0.0035	0.0004	–
2×	Eligible workers	0.1918	0.2050	0.0131	6.8
	Whole population	−0.0102	−0.0093	0.0010	–
Concentrated cities					
1×	Eligible workers	0.1621	0.1822	0.0201	12.4
	Whole population	0.0030	0.0047	0.0018	–
2×	Eligible workers	0.4252	0.4766	0.0514	12.1
	Whole population	0.0038	0.0084	0.0045	–

Notes: Welfare gains in percent of consumption of goods and housing, averaged over the group shown with benchmark population weights; shift-share calibration, 2019. Eligible workers are those whose benchmark cell wage is below 60 percent of the employment-weighted median wage (3.8 percent of employment). Establishments are held at their benchmark values, since free entry has no interior equilibrium in some markets under this calibration. The difference is the column with labor market power less the column without, in percentage points and as a percent of the latter. The percent is omitted for the whole population under the subsidy, where the gain without labor market power is close to zero.

the entry-fixed column of Table 4, not to its free-entry column. With labor market power, the welfare gain of the housing policy is 0.4 percent larger than without it when the policy targets New York, San Francisco, and San Jose and 2.8 percent smaller when it targets the concentrated cities, and the gain of the migration subsidy to eligible workers is 6.9 and 12.4 percent larger. The subsidy's effect on the whole population has the same sign in both models, as it does under the main calibration when the number of establishments is held fixed.

E Markdown Decomposition in General Equilibrium

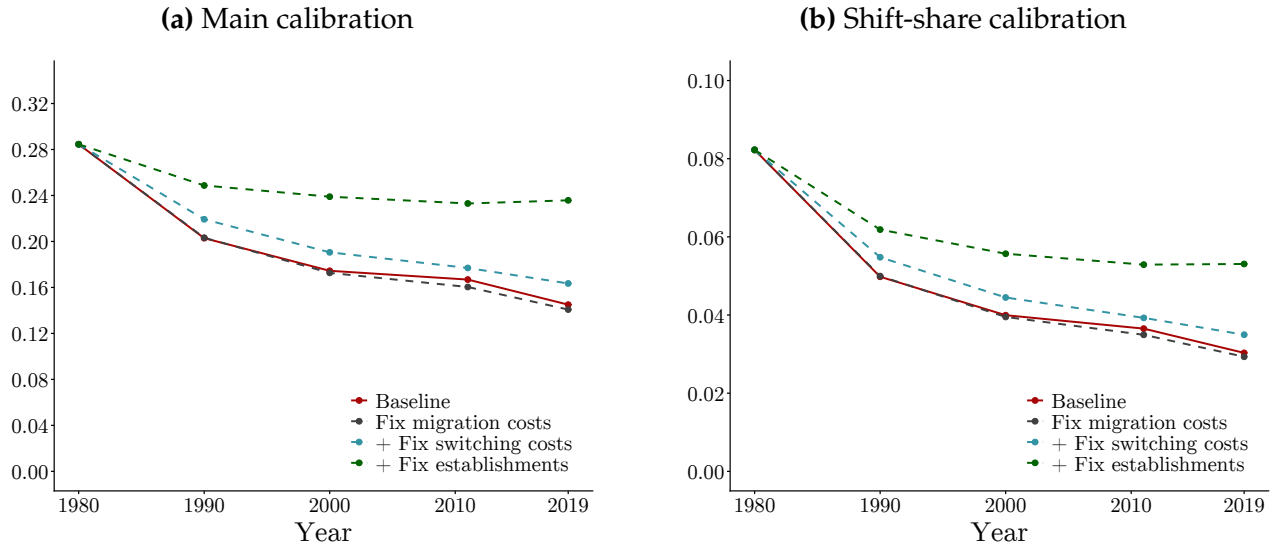
In contrast to the partial-equilibrium design of Section 5, this appendix reports a general-equilibrium variant of the markdown decomposition in which unobserved utilities are held at their benchmark values while wages, rents, populations, and the agglomeration-driven productivity shifters adjust endogenously under the same counterfactual configurations of migration costs, switching costs, and establishment counts. Whenever establishment counts

employment in the concentrated cities.

are not held fixed, they adjust through the free-entry condition, so entry itself becomes an equilibrium response.

Figure E1 shows the resulting markdown trajectories under the main and shift-share calibrations; Table 3 in the main text reports the Shapley attributions for all four specifications, and Table E1 their year-by-year evolution under the main calibration.

Figure E1: Mean markdowns across location-industries in general equilibrium



Notes: Conventions as in Figure 2, with wages, rents, populations, establishment counts, and agglomeration-driven productivities adjusting endogenously under each counterfactual configuration and unobserved utilities held at benchmark values. In 1980 all series coincide with the partial-equilibrium value by construction.

The channels rank consistently across all four specifications. Industry-switching costs and establishment counts each account for a positive share of the observed markdown decline, and migration costs for a negative share. The general-equilibrium variant lowers the explained share relative to partial equilibrium, from 71 to 65 percent under the main calibration, and the reduction is concentrated in the switching channel, from 36 to 28 percent, while the establishment channel is unchanged at 45 percent. Under free entry, 1980 switching costs raise markdowns and hence per-establishment operating profits, so the free-entry condition brings additional establishments into operation. These establishments compete for workers and push markdowns partly back down. Under the shift-share calibration the explained share falls by more, from 57 to 44 percent, and both the switching channel, from 31 to 20 percent, and the establishment channel, from 34 to 29 percent, shrink.

The migration channel's Shapley share is small and positive in 1990, turns negative afterwards, and is largest in absolute value in 2011 (Table E1), consistent with estimated migration costs rising mostly after 1990.

Table E1: Year-by-year contribution to the decline since 1980 (%), main calibration

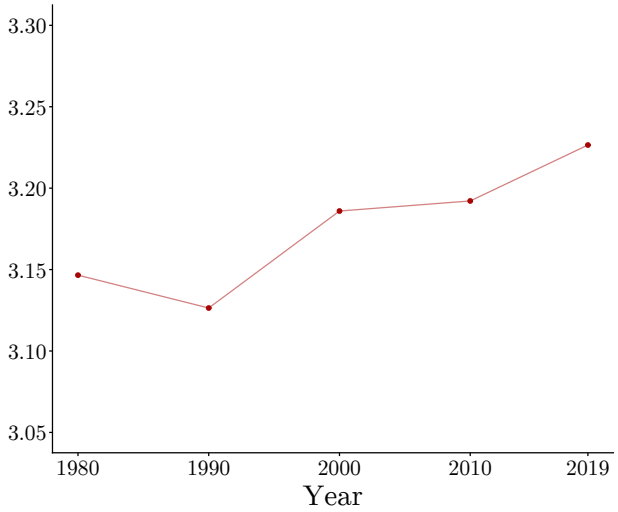
	Partial equilibrium				General equilibrium			
	Establishments	Switching	Migration	Explained	Establishments	Switching	Migration	Explained
1990	27	35	1	63	28	27	1	56
2000	38	31	−4	65	38	24	−3	59
2011	47	30	−15	62	45	23	−11	56
2019	45	36	−10	71	45	28	−8	65

Notes: Channel columns report Shapley shares, in percent of the total decline in the employment-weighted mean markdown between 1980 and the indicated year; in each year the three shares sum to the explained share, up to rounding. The 1980 row is identically zero by construction and omitted. Main calibration in both panels.

F Additional Figures and Tables

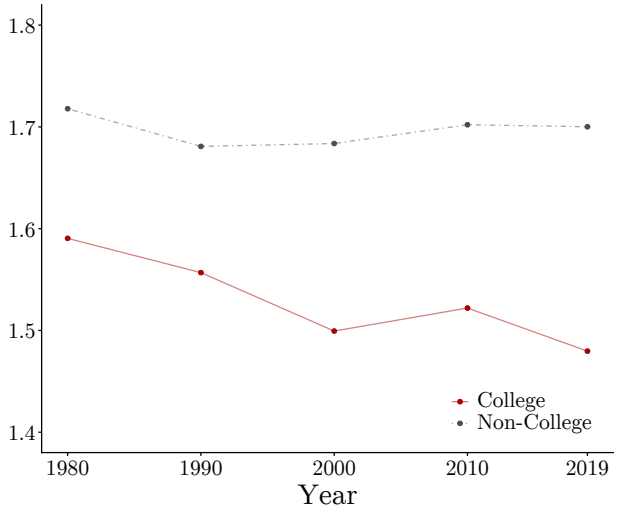
This appendix collects the supplementary figures and tables referenced in the main text. Figure F1 plots the mean number of symmetric establishments per local labor market over time. Figure F2 plots the mean task distance between industries, the measure that defines the switching cost bins. Figure F3 reports the five-year industry switching rate by city size. Figure F4 shows the estimated cumulative migration cost schedules for each sample year; Tables F1 and F2 report the estimated migration and industry-switching costs by year and education group, and Table F3 summarizes the calibrated location-industry parameters under the main calibration.

Figure F1: Symmetric establishments per market, log mean



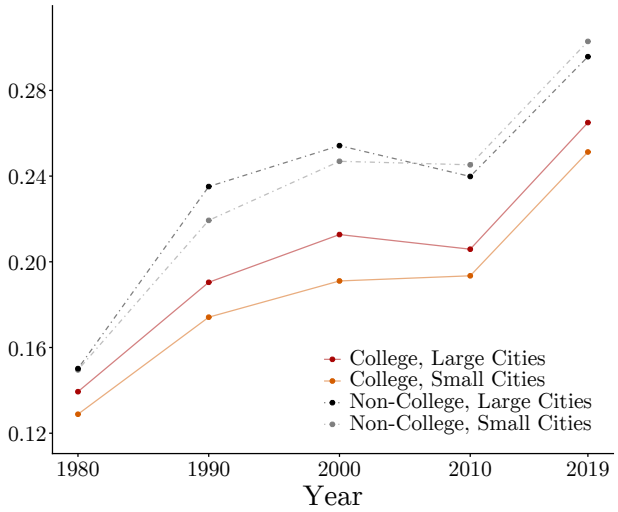
Notes: CBP data (1980–2016). The plotted series is the log of the unweighted mean number of symmetric establishments across location-industry cells. "Symmetric" establishments refer to the hypothetical number of equally-sized establishments that would yield the observed local employment concentration (Adelman 1969).

Figure F2: Mean task distance across industries



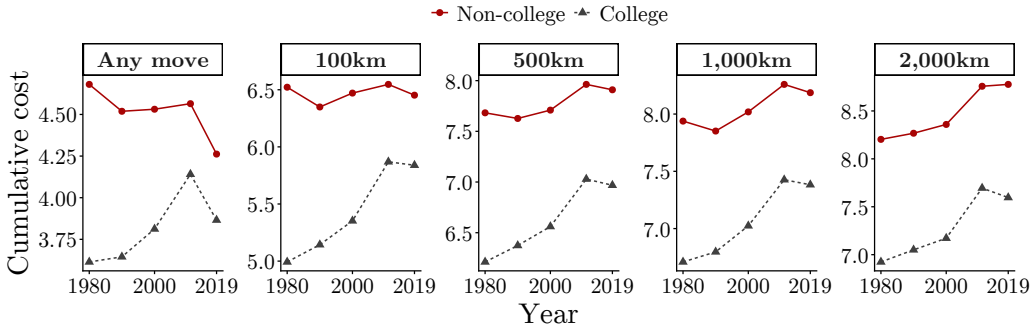
Notes: Mean task distance between industries by education group, computed as the unweighted mean of the pairwise Euclidean distances between industries' task vectors in each sample year. Task vectors are constructed from the 1977 Dictionary of Occupational Titles following Autor and Dorn (2013).

Figure F3: Five-year industry switching rate by education and city size



Notes: CPS data (1975–2017, pooled around each sample year, with 2015–2017 for 2019). Each series is the share of workers who change 3-digit Census industry over a five-year horizon, by education group and city size; large cities are those in the top quartile of city population in each sample year. The CPS records one-year transitions, which are converted to five-year equivalents as described in Appendix B.1.

Figure F4: Migration cost estimates across sample years



Notes: Cumulative moving cost at each distance threshold, computed as the sum of the estimated cost increments in equation (28) up to that threshold, re-estimated separately for each year and education group. Costs are in utility units.

Table F1: Utility cost of moving

Year	Education	Any move	100km	500km	1,000km	2,000km
1980	College	3.614	1.378	1.219	0.497	0.216
1990	College	3.645	1.499	1.230	0.423	0.254
2000	College	3.812	1.540	1.208	0.463	0.148
2011	College	4.141	1.728	1.160	0.397	0.269
2019	College	3.866	1.974	1.129	0.415	0.213
1980	Non-college	4.681	1.842	1.160	0.256	0.265
1990	Non-college	4.519	1.830	1.278	0.225	0.416
2000	Non-college	4.531	1.940	1.238	0.310	0.340
2011	Non-college	4.565	1.982	1.416	0.296	0.498
2019	Non-college	4.262	2.191	1.457	0.277	0.590

Notes: The table shows parameter estimates of the moving cost function specified in equation (28). The first column reports the fixed utility cost of any move; each remaining column reports the utility cost increment for moves above the stated distance.

Table F2: Utility cost of switching industries

		Utility cost increment for task distance above					
Year	Education	0.05	0.3	0.8	1.5	2	4
1980	College	6.311	0.045	−0.011	−0.001	0.259	0.274
1990	College	5.660	0.249	0.127	0.093	0.085	0.586
2000	College	5.407	0.393	0.045	0.111	0.200	0.347
2011	College	5.407	0.320	0.065	0.178	0.133	0.485
2019	College	4.981	0.271	0.064	0.319	0.145	0.421
1980	Non-college	6.602	0.129	−0.125	0.055	−0.168	0.549
1990	Non-college	5.985	0.046	−0.028	0.037	0.087	0.467
2000	Non-college	5.839	−0.132	0.104	0.091	0.084	0.602
2011	Non-college	5.770	−0.071	0.061	0.119	0.116	0.419
2019	Non-college	5.354	−0.046	0.033	0.174	0.187	0.372

Notes: The table shows parameter estimates of the switching cost function specified in equation (30). Each parameter represents the utility cost increment for task distances above the specified thresholds.

Table F3: Summary statistics of calibrated parameters

	Observations	Mean	SD	Min	Max
Labor Supply					
Unobserved amenities Non-college ξ^n 1980	15,122	28.720	1.621	23.586	34.837
Unobserved amenities College ξ^c 1980	15,122	148.917	1.709	143.943	156.552
Unobserved amenities Non-college ξ^n 2019	25,032	26.553	2.285	18.641	35.035
Unobserved amenities College ξ^c 2019	25,032	217.479	2.502	208.459	226.988
Labor Demand					
TFP parameters A_{jkt} 1980	15,122	183899.428	139068.575	35469.616	2,668,019.984
TFP parameters A_{jkt} 2019	25,032	145100.402	88953.459	27624.084	1,875,744.425
Agglomeration constant a_{jkt} 1980	15,122	12.622	0.364	11.828	13.763
Agglomeration constant a_{jkt} 2019	25,032	12.465	0.371	11.670	13.605
Factor Non-college θ^n 1980	15,122	0.607	0.158	0.049	0.963
Factor College θ^c 1980	15,122	0.393	0.158	0.037	0.951
Factor Non-college θ^n 2019	25,032	0.498	0.173	0.019	0.985
Factor College θ^c 2019	25,032	0.502	0.173	0.015	0.981
Housing Supply					
κ_{jt} 1980	15,122	181.639	384.028	0.000	3,553.006
κ_{jt} 2019	25,032	165.999	351.705	0.000	3,494.568

Notes: Summary statistics of the internally calibrated parameters across location-industry cells, main calibration, for the first and last sample years. Factor shares satisfy $\theta^n_{jkt} + \theta^c_{jkt} = 1$.